MONITORING, MODELLING APPROACHES AND EVALUATION OF THE RESILIENCE OF BIORETENTION SYSTEMS

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Abstract :

The imperviousness of urban areas inhibits processes such as infiltration and evapotranspiration which allow stormwater to re-enter the natural water cycle. As a result, the demand on existing sewage systems is increasing due to excessive amounts of stormwater runoff. The upsurge in storm frequency and intensity is only exacerbating this demand. Bioretention systems, a Nature-Based Solution (NBS) for stormwater management, allow for the passive capture and treatment of stormwater. Monitoring and modelling of such systems is important to ensure the success of their design, implementation, and maintenance. In this study, a review of existing literature in the domain of monitoring and modelling of bioretention systems in the context of climate change is presented, four case studies of bioretention monitoring in different climates and configurations were cultivated be used to validate models' ability to simulate performance in various climates, and a method for assessing the resilience of bioretention systems was adapted. Preliminary validation of newly developed infiltration models using data from one case studies is shown, as well.

Key words : nature-based solutions, climate change, stormwater management, climate change, urban hydrology, infiltration, field validation

1. INTRODUCTION

Urban growth and the resulting increase in impervious surfaces have had a significant impact on the natural hydrological cycle, particularly under climate change with heavier and more frequent precipitations (Seneviratne et al., 2021). This interference has created a pressing need for innovative stormwater management solutions. Impervious surfaces, such as roads, rooftops, and parking lots, disrupt the natural processes of infiltration (where water seeps into the ground) and evapotranspiration (ET, the combined process of water evaporation from the soil and transpiration from plants). Consequently, several issues arise.

Among these issues, one can mention the urban heat island effect, which is characterized by higher temperatures in cities compared to surrounding rural areas. The lack of infiltration and ET reduces the cooling effects that vegetation and natural surfaces provide, leading to increased

heat retention in urban environments. This phenomenon not only causes discomfort for residents, but has broader implications for energy consumption and public health (Tong et al., 2021). Decreased infiltration also impacts groundwater resources, since less water is able to recharge groundwater aquifers. Coupled with the greater water demand experienced by urban areas, this can lead to depletion of groundwater resources.

The alteration of the natural water cycle due to urbanization also contributes to generation of significant amounts of stormwater runoff. With limited opportunities for absorption into the ground, rainfall quickly accumulates on impervious surfaces, creating excess runoff. In high quantities, this runoff poses a considerable challenge in the domain of urban stormwater management, as it overwhelms existing drainage systems and increases vulnerability to flooding.

Excess runoff has also an effect on receiving water bodies. From a hydrological perspective, the increased quantity of inflow creates changes in the timing, duration, and intensity of peak flows in rivers and streams (Jacobson, 2011; Seneviratne et al., 2021). Urbanization also contributes to water quality degradation, since stormwater runoff collects various pollutants – such as heavy metals, nutrients, and organic waste – as it comes into contact with impervious surfaces. When this contaminated runoff reaches larger receiving water bodies, serious environmental and sanitary risks are posed (W. Ahmed et al., 2019; Pascaud et al., 2015; Sidhu et al., 2013).

To address these challenges, bioretention has emerged as a component of many nature-based solution (NBS) for sustainable stormwater management, such as bioswales, infiltration basins, and rain gardens. These solutions are known by various names globally, such as Sustainable Stormwater Drainage Systems (SUDS) in Australia and the UK, Low Impact Development (LID) or Best Management Practices (BMP) in North America, and Techniques Alternatives

(TA) in France (T. D. Fletcher et al., 2015). Regardless of its terminology, these interventions aim to augment the sustainability and resilience of existing stormwater management infrastructures, thus mitigating the detrimental effects of urbanization.

Bioretention systems typically consist of specially designed vegetated areas – often integrating engineered structures such as underdrains – that capture and passively treat stormwater. Many different configurations of such systems exist, and are chosen according to the specific needs of the sites at hand. Through the creation of conditions suitable for infiltration and ET to occur, they are able to re-implement elements of the natural water cycle in urban areas, and counteract the aforementioned repercussions of increased urbanization. As runoff flows into bioretention areas, vegetation and soil help filter out pollutants and retain water. The treated water can then slowly infiltrate into the ground, recharging aquifers, or be released at a controlled rate, reducing the strain on conventional drainage systems and decreasing the risk of flooding. The term "bioretention" is based on the way that these systems use biological components such as soil and vegetation for the retention of water and pollutants.

The incorporation of bioretention systems into urban infrastructure is becoming increasingly relevant especially in the context of climate change. With the increase in extreme weather events (Seneviratne et al., 2021) current urban stormwater infrastructure will be subjected to new demands (Berggren et al., 2012; S. Fletcher et al., 2019; Rosenberg et al., 2010; Semadeni-Davies et al., 2008; Willems et al., 2012). Increased storm frequency and intensity will need to be accounted for in order to maintain the integrity of urban areas. In Romania, Pânzaru et al. (2022) found evidence of urbanization's impact on urban water availability and quality, and recommends NBS water management to be implemented in conjunction with conventional engineering to increase climate resilience. Adu Boateng et al. (2023) also found that there is a

need for such measures to be implemented into urban development in Kumasi, Ghana, for climate change mitigation and adaptation.

For successful design, implementation, and maintenance of bioretention systems, monitoring and modelling are essential. Monitoring consists of regular data collection and analysis of different parameters such as infiltration and pollutant removal efficiency, whereas modeling makes use of models to simulate a system's performance under different situations. For the development of models, monitoring data is used to validate the model's ability to accurately represent a system's behavior. There are many existing models for bioretention design, although there is a challenge in finding a balance between accuracy and facilitated operation – many models are highly complex involving many parameters and long computation times.

Therefore, there is a need for accurate and easy-to-operate hydrological models. In order to account for the effects of climate change and assess the resilience of existing bioretention systems, it is crucial that these models have the capacity to accurately represent the behavior and performance of such systems across varying weather regimes. Furthermore, the validation and calibration of these models using diverse field hydrological monitoring data is essential.

The main objectives of this study are to present a review of current research in bioretention modelling and monitoring, prepare case studies of bioretention systems with field data from different climates and with different configurations to be used for the evaluation of models, and to develop a method for testing the resilience of the systems.

2. METHODS

2.1 Literature Review

A review of current literature in the domain of modelling and monitoring of bioretention was conducted. A total of 86 articles were reviewed.

The different configurations of bioretention systems were explored, as well as the various methods and instrumentation used to monitor them, including various monitoring studies. Additionally, the dominant models presently used to simulate these systems were investigated through the parameters and equations involved in their function, as well as past validation studies. An overview of their respective strengths and shortcomings was also provided. Furthermore, the impacts of climate change on urban drainage systems were researched, the concept of resilience was defined, and methods of resilience assessment were analysed.

2.2 Data Acquisition

Through the literature review, several case studies of bioretention monitoring were found, four of which were selected for the purpose of this study. The monitoring studies were chosen based on the following criteria: the amount of information or on the testing site (at least needed to include the area of the cell, catchment area, description of the media), the kind of data recorded (flow rates, water level, storage, precipitation), type of study (field as opposed to pilot-scale). Additionally, the sites were to have diverse configurations and climate classifications. Four cases, located in: Omaha, USA; São Carlos, Brazil; Singapore; and Vaughan, Canada; met these criteria and had supplemental information including daily summaries of water balance data available to the public.

The authors of each of the four studies were contacted via email in regards to accessing data with high temporal resolution (between 1- and 6-minute timesteps). Responses from authors were received for three of the four studies, although only two (Batalini de Macedo et al., 2019; J. Wang et al., 2019) were able to provide data that met the temporal resolution criteria aforementioned.

The behaviours of the two sites with high temporal resolution data were compared in order to observe the influence of system configuration under similar rainfall conditions. In order to do

this, data from events from the São Carlos, Brazil, site with the most similar total rainfall depths to the data from Singapore were chosen (Table 1). Comparisons were made on the basis of overflow and surface storage volume.

Table 1. Dates and total rainfall depths of the events chosen to compare the functionality of the Singapore and Brazil bioretention sites.

Singapore		Brazil	
Date	Total rainfall depth (mm)	Date	Total rainfall depth (mm)
26/06/2013	18.8	24/04/2017	18.6
09/07/2013	41.4	05/05/2017	10
17/07/2013	11	21/05/2017	23.4
11/08/2013	39.2	14/02/2018	38.4
13/08/2013	11.6	19/02/2018	12.4
28/08/2013	9.8		
05/09/2013	27.6		

2.2 Modelling

2.2.1 Presentation of models to be validated

The data selected was intended to be used for the validation of the three physically based models which attempt to maintain physical soundness while avoiding overcomplication. These models take into account factors such as hydric condition, soil parameters, and geometric features to describe the water flux using hydraulic gradient and conductivity. They also assume that there is no water flux at the base of the soil block. The first two models, CH1 and CH2, take into account the increase in hydraulic conductivity with the amount of water infiltrated, whereas the third, CH3, only considers the saturated hydraulic conductivity (Asry et al., 2023).

The parameters considered are: saturated hydraulic conductivity, shape parameters n and l, porosity, water content, pressure head, and depth of soil. The CH1 model computes the equivalent hydraulic conductivity as a function of cumulative infiltration, $K_{eq,CH1}$(I), based on the unsaturated hydraulic conductivity using:

$$K_{eq,CH1}(I) = K_s\left(S_{e,i} + \frac{I}{z_{soil}(\theta_s\theta_r)}\right)^l\left(1-\left(1-\left(S_{e,i}+\frac{I}{z_{soil}(\theta_s\theta_r)}\right)^{\frac{1}{m}}\right)^m\right)^2 \quad (7)$$

where K_s is the saturated hydraulic conductivity, I is the cumulative infiltration, $S_{e,I}$ is the degree of saturation, z_{soil} is the soil block depth , θ_s is the saturated water content, θ_r is the residual water content, l and m (Mualem's condition) are hydraulic shape parameters. The hydraulic gradient, $i_{eq,CH1}$(I), is computed using:

$$i_{eq,CH1}(I) = 1 + \left(\frac{h_a}{z_{soil}} - \frac{\theta_i}{\theta_s} - \frac{I}{z_{soil}\theta_s}\right) \quad (8)$$

where h_a is the water pressure head imposed at the surface and θ_i is the initial water content.

CH2 uses the same method for the equivalent hydraulic conductivity, $K_{eq,CH2}(I)$, as CH1 (eq 1), but takes into account only the change in hydraulic gradient, $i_{eq,CH2}$, using :

$$i_{eq,CH2-3}(I) = 1 + \frac{\left[h_a + z_{soil}\left(\frac{\theta_s-\theta_i}{\theta_s}\right)\right](\theta_s-\theta_i)}{l} \quad (9)$$

CH3 uses the same equation as CH2 (eq 3) to compute the change in hydraulic gradient, $i_{eq,CH3}$, while the equivalent hydraulic conductivity, $K_{eq,CH3}(I)$, is fixed at the saturated hydraulic conductivity, corresponding to a version of the Green-Ampt model. A more detailed overview of the models and their derivations can be found in Asry et al. (2023).

2.2.2 Modelling process

The previously mentioned models, CH1-3, were meant to be validated using the CANOE software. Due to delays in uploading the newest models to the software, experimentation with a previous version took place. See Figure 1. Modelling the components of the Wicks Reservoir using an earlier version of CANOE (Bonneau et al. 2020)for a representation of the components of a bioretention cell using in CANOE before the integration of CH1-3.

The code for model CH1 in R was used to simulate seven events from the case study in Singapore (J. Wang et al., 2016). Performance was evaluated on the basis of Nash Sutcliffe Efficiency, PBIAS, and RMSE.

2.3 Resilience Evaluation

A method for assessing the resilience of bioretention systems was developed, incorporating an adaptation tipping point approach: the magnitude of a climate change effect at which a system's capacity is reached. More specifically, this means of evaluation was adapted from the mainstreaming method developed by Gersonius et al. (2012) represented in Figure 7.

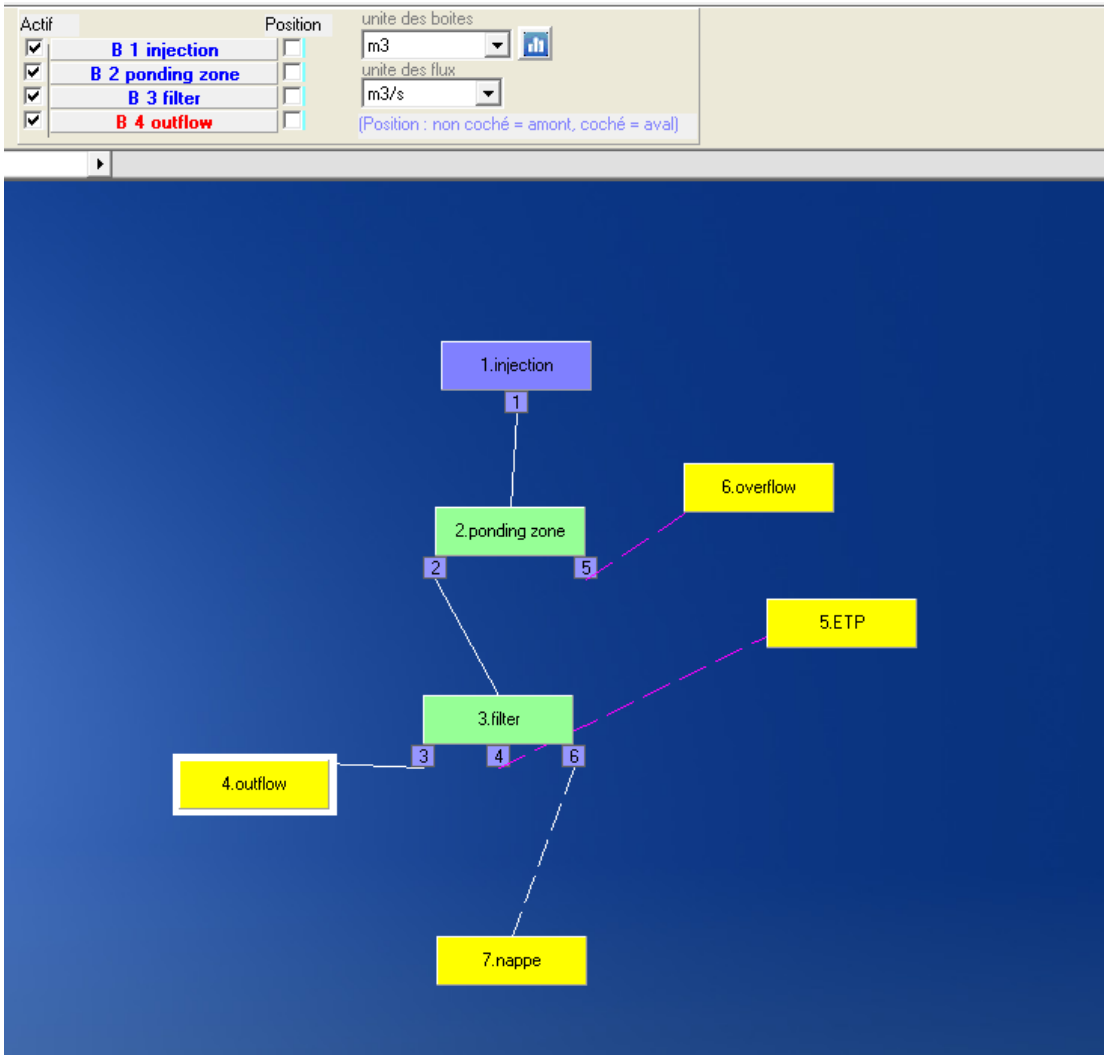

Figure 1. Modelling the components of the Wicks Reservoir using an earlier version of CANOE (Bonneau et al. 2020)

3. LITERATURE REVIEW: MONITORING AND MODELLING OF BIORETENTION SYSTEMS IN A CHANGING CLIMATE

3.1 Bioretention design

Urban drainage is demanding more attention in light of cities' increased vulnerability to flooding and infrastructure failure due to climate change and its impact on precipitation patterns. Bioretention systems are a form of low impact development for stormwater management used to attenuate peak flows, decrease stormwater volume, and remove pollutants. They incorporate the passive treatment of stormwater in which runoff flows into a cell before

percolating through a soil filter after which it is infiltrated or discharged (Un, 2016). The cells typically consist of a ponding zone with vegetation at the surface, multiple layers of media for filtration and storage, and a drainpipe at the bottom for discharge into a reservoir, etc.

Design specifications for such systems are demanding revision due to the need to accommodate new climatic conditions.

Melbourne Water identified five main bioretention system configurations (Melbourne Water, 2020). A pipeless system (Figure 2) has the simplest design, incorporating neither an underdrain nor a raised outlet. Additionally, it lacks a base liner, only a permeable liner on the sides permitting exfiltration into the surrounding soil. Both temporary (Figure 3) and longer lasting (Figure 4) submerged zone configurations also lack underdrains but consist of raised outlets which create internal water storage (IWS) for vegetation and fauna. The temporary type consists of a permeable side liner and no base liner, whereas the longer lasting type has impervious liner on the sides as well as the base for longer-term IWS. Basic (Figure 5) and contained (Figure 6) systems both have underdrains for drainage when moisture uptake and soil infiltration capacity is exceeded but lack raised outlets. Similar to the temporary submerged zone, the basic system has a permeable side liner and no base liner for the purpose of exfiltration, while the contained system, like the longer lasting submerged zone, has impermeable liner on the sides and the base.

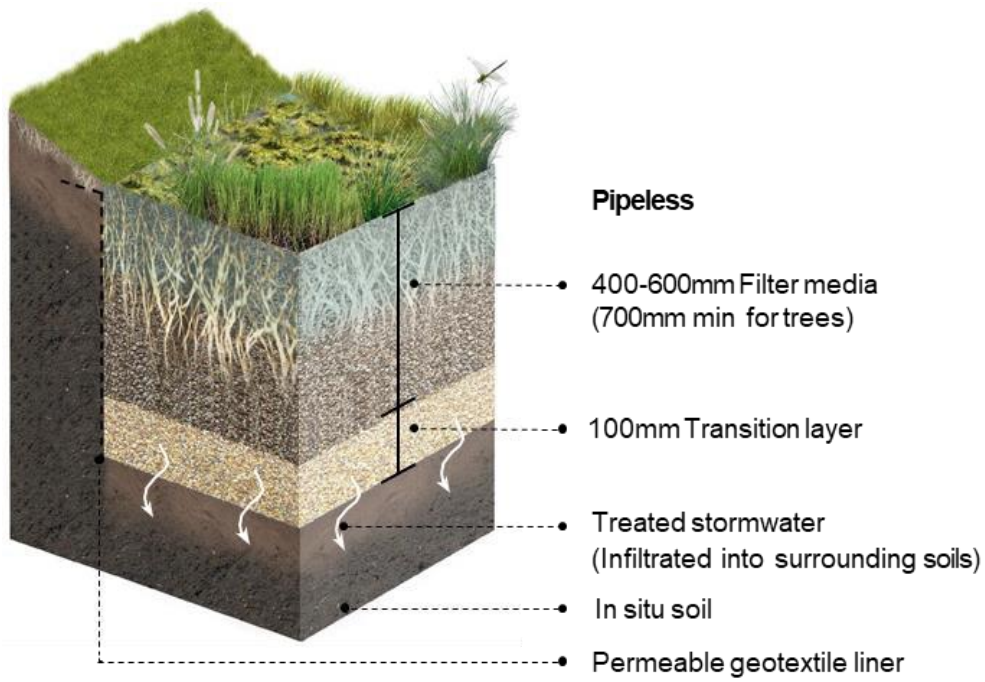

Figure 2. Pipeless bioretention system (Melbourne Water 2020)

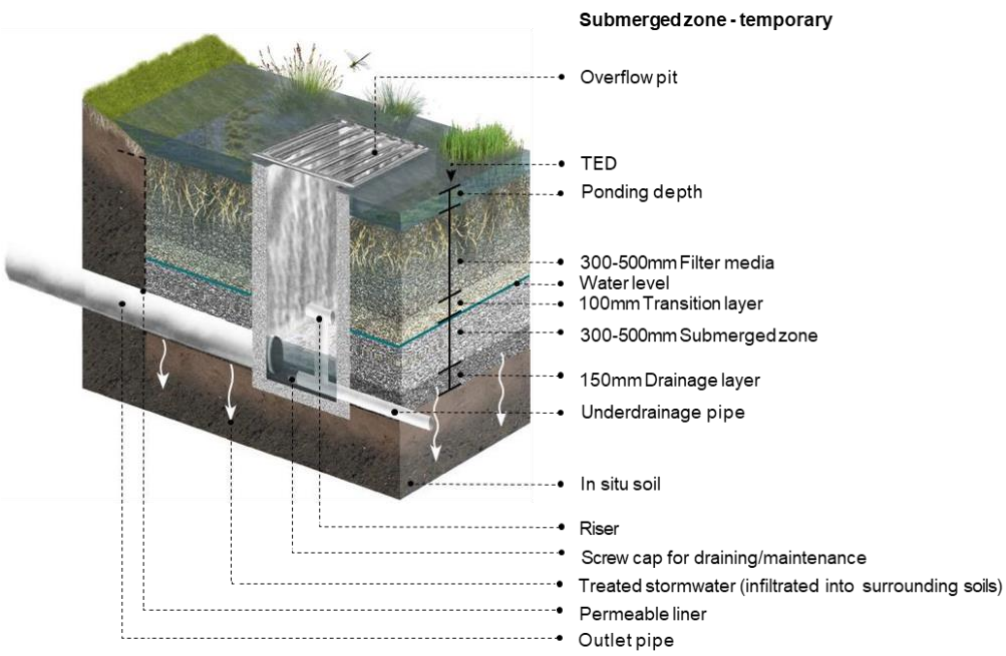

Figure 3. Bioretention system with a temporary submerged zone (Melbourne Water 2020)

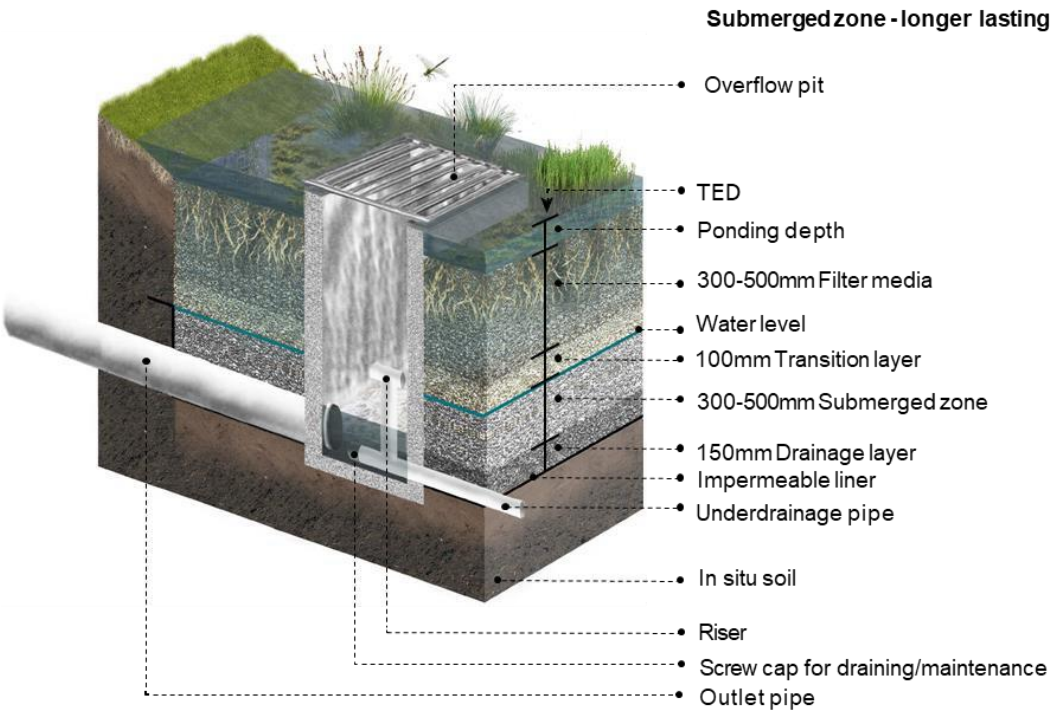

Figure 4. Bioretention system with a longer lasting submerged zone (Melbourne Water 2020)

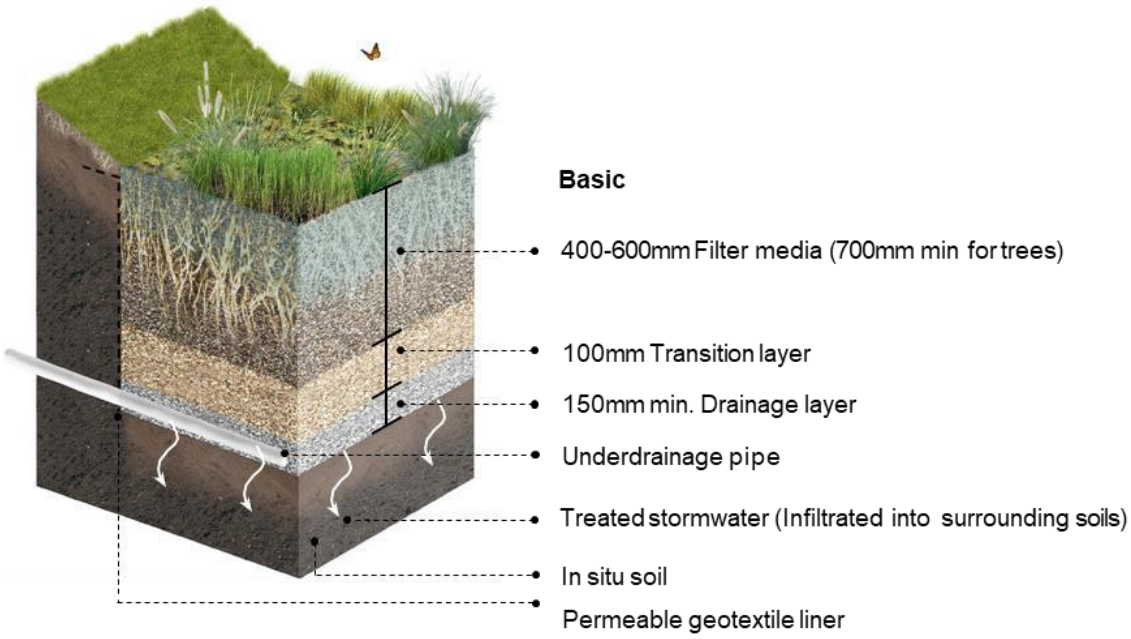

Figure 5. Basic bioretention system (Melbourne 2020)

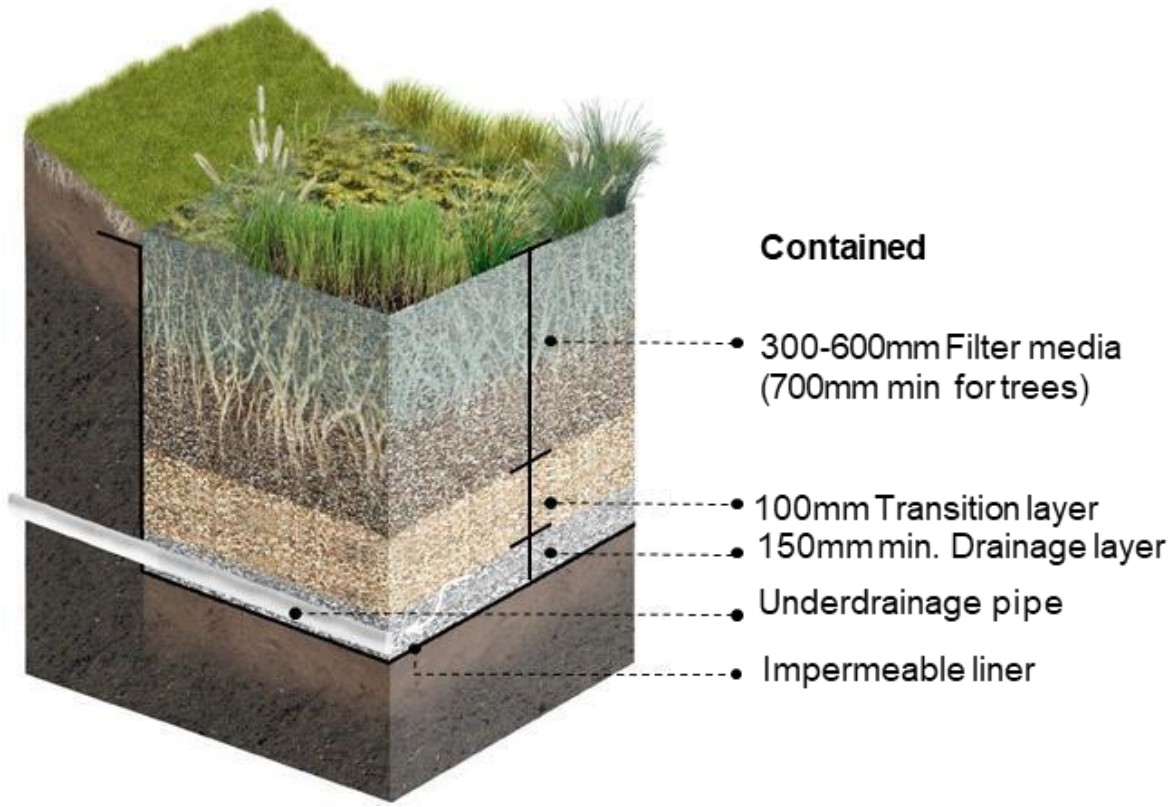

Figure 6. Contained bioretention system (Melbourne Water 2020)

Bioretention design depends greatly on stormwater runoff properties, site requirements, geographic location, and local availability of constituents. The type of media is especially critical to hydrologic efficiency; substrate water holding is important during drought conditions and delays peak flows (Vijayaraghavan et al., 2021). Inflow into the bioretention cell depends on rainfall duration and intensity, whereas outflow depends on ponding depth (the area reserved for water storage before it is infiltrated) and the saturation of the media (Olszewski & Davis, 2013). The primary factors dictating whether outflow (drainage and overflow) will occur are: the hydraulic conductivity of the soil, IWS depth, and the surface infiltration rate. Pilot-scale mesocosm studies are often conducted to examine performance. In the review, Vijayaraghavan et al. (2011), of bioretention for stormwater management, some of their recommendations for future research include long term studies, optimization of individual components (such as

vegetation, soil media, and drainage), as well as flexibility for different climates and predictive capability of models.

3.2 Monitoring

There are many aspects of bioretention system functionality, and therefore many approaches for their monitoring. From water quality and hydrological function, to infiltration and soil characteristics, the breadth is large.

Water balance, which involves the tracking and quantification of the movement and distribution of water within a bioretention system, is an important measure of hydrological performance. Some of the different parameters involved include: precipitation, inflow and outflow rates, ET, infiltration rate, storage volume, and water level.

Although precipitation is not included in inflow measurements, it may constitute a significant portion of the water entering a bioretention system in some cases. Therefore, rainfall data must be measured and accounted for. Some of the tools used include depth measurement rain gauges, accumulating rain gauges and tipping bucket rain gauges. Tipping bucket rain gauges are considered to be the most accurate since they involve continuous recording via data logger, and are therefore recommended for timely measurements of depth and intensity (Anderson et al., 2008). For this reason, they are commonly utilized in scientific studies (Bonneau et al., 2020; Fowdar et al., 2022; Van Seters & Graham, 2014; J. Wang et al., 2016).

There are a variety of measurement devices available for measuring flow and water level. According to the Urban Stormwater BMP Performance Monitoring: A Guidance Manual for Meeting the National Stormwater BMP Database Requirements (2002), primary flow measurement devices include weirs (Batalini de Macedo et al., 2019; Bonneau et al., 2021a)

and flumes mounted by data loggers to measure discharge and are used in conjunction with secondary flow measurement devices used to measure water depth. The guide specifies that weirs are easier to install and less expensive, but flumes tend to be more accurate and require less maintenance. Some secondary flow measurement tools include water level probes (Bonneau et al., 2020) and pressure transducers (Stewart et al., 2017; J. Wang et al., 2016) as well as float gauges and ultrasonic depth sensors, among others.

In the context of bioretention, ET is the combination of water vapor evaporated from the soil and transpired by vegetation. The most commonly used tool for direct measurement is the Lysimeter, which consists of a soil-filled tank in which plants are grown and measures the water lost by transpiration and evaporation (*Measurement and Estimation of ETc*, 2012). ET can also be estimated using theoretical models. The American Society of Civil Engineers has developed a widely used standardized reference evapotranspiration equation based on the Penman-Monteith equation which uses the principles of energy balance and mass transfer (Walter et al., 2005).

3.3 Infiltration

Infiltration rate measurements are also a means to evaluate the efficiency of a bioretention system. There are different types of infiltrometers, although the most common is the double ring infiltrometer (DRI), which consists of two concentric cylinders. ASTM International uses DRI in their procedure for measuring infiltration (ASTM-D3385). Another common infiltrometer uses a single ring (SRI). When comparing DRI and SRI, Verbist et al. (2010) found no significant differences between the two instruments for most calculation methods. However, Gregory et al. 2005 claims that SRI often overestimate infiltration rates, an error minimized by DRI since the outer ring forces vertical infiltration of water in the inner ring and reduces potential leakage. Philips and Kitch (2011) describe various direct and indirect in-situ

tests such as the cone penetration test, borehole infiltration test, and the pilot infiltration test. They mentioned that the DRI test is best suited for soils with low hydraulic conductivity. In a study examining the infiltration of different porous materials, Valinski & Chandler (2015) chose to use an automated mini-disk infiltrometer (AMDI) instead of a ring infiltrometer. The study claimed that, compared to AMDI, ring infiltrometers have a significant set up time, limit the number of experimental replicates (due to fixed-nature), provide a poor understanding of special variability and are not always representative of depth of water associated with certain rain events. In this case, it was found that the ADMI provided comparable infiltration rates to a ring infiltrometer, although it would not be compatible for testing infiltration where substantial ponding depths are expected (Valinski & Chandler, 2015). Soil moisture can also be used to determine infiltration and is often measured using time domain reflectometers (F. Ahmed & Borst, 2019).

To address the shortcomings aspects of traditional methods for studying soil hydraulic processes, Lassabatère et al. (2006) developed the Beerkan estimation of soil transfer parameters (BEST). To determine the water retention curve, it takes into account the van Genuchten relation (Eq. 1a) with the Burdine condition (Eq. 1b) and the Brooks and Corey relation for hydraulic conductivity (Eq. 2):

$$\frac{\theta-\theta_r}{\theta_s-\theta_r} = \left[1+\left(\frac{h}{h_g}\right)^n\right]^{-m} \qquad (1a)$$

$$m = 1-\frac{2}{n} \qquad (1b)$$

$$\frac{K(\theta)}{K_s} = \left(\frac{\theta-\theta_r}{\theta_s-\theta_r}\right)^{\eta} \qquad (2)$$

Where n, m, and η are shape parameters and θ_s, θ_r, and K_s are scale parameters. BEST was found to be a promising method for characterizing the hydraulic behavior in terms of reliability, cost-effectiveness and ease of use (Lassabatère et al., 2006).

3.4 Modelling

Various software exists for bioretention modelling. Lisenbee et al. (2021) reviewed seventeen models capable of modelling bioretention cell hydrology and found HYDRUS and GIFMod to be the only models capable of determining infiltration under variably saturated conditions. Both were also the only models in the review found to use Richard's equation (Eq. 3c) for infiltration at the soil surface. Richard's equation (Richards, 1931) is considered to be most comprehensive infiltration model, combining Darcy's law (3a) and the mass conservation equation (Eq. 3b). However, it requires two soil properties that are not easily measured: unsaturated hydraulic conductivity curve and the soil water characteristic curve (SWCC).

$$q = -K(\theta)\left(1 + \frac{\partial h}{\partial z}\right) \qquad (3a)$$

$$\frac{\partial \theta}{\partial t} = -\frac{\partial q}{\partial z} \qquad (3b)$$

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left[K(\theta)\left\{\frac{\partial h}{\partial z} + 1\right\}\right] - S(\theta) \qquad (3c)$$

Where q [LT^{-1}] is the water flux, θ [L^3L^{-3}] is the soil volumetric moisture content, h [L] is the suction head, z [L] is vertical position, t [T] is time, $K(\theta)$ [LT^{-1}] is unsaturated hydraulic conductivity, S [T^{-1}] is the plant transpiration rate and $\partial\theta/\partial h$ is the soil moisture capacity function, with the assumption of one-phase, vertical matrix flow with isothermal conditions and no air effects (Dussaillant-Jones et al., 2004).

Furthermore, GIFMod (Massoudieh et al., 2017), a relatively new model, uses Darcy's law to model saturated infiltration between layers and the van Genuchten equation (van Genuchten, 1980) with the Mualem condition (Mualem, 1976)for unsaturated infiltration to be used in the Richards equation. The Mualem-van Genuchten equation (Eq. 4) is most often used to describe find the unsaturated soil hydraulic conductivity from measured saturated hydraulic conductivity and the soil water retention curve (SWRC) (Tian et al., 2019).

$$S_e = \frac{(\theta-\theta_r)}{(\theta_s-\theta_r)} = (1+(\alpha h)^n)^{-m} \qquad (4a)$$

$$K(S_e) = K_s S_e^{\tau}\left(1-\left(1-S_e^{\frac{1}{m}}\right)^m\right)^2 \qquad (4b)$$

Where S_e is the saturation degree, θ [L^3L^{-3}] is the volumetric water content, θ_r [L^3L^{-3}] and θ_s [L^3L^{-3}] are the residual and saturated water contents, α [L^{-1}] is the scale parameter for water pressure head h, m and τ are shape parameters, K is the unsaturated hydraulic conductivity [LT^{-1}], K_s [LT^{-1}] is the saturated hydraulic conductivity (Bonneau et al., 2021b).

Massoudieh et al. (2017) demonstrated GIFMod's effectiveness and flexibility in representing processes affecting flow as well as pollution transport and transformation in green infrastructure systems. However, the review, Kaykahosravi et al. (2018), mentions that GIFMod has limitations when considering plant-related processes involved in the water balance, such as root water uptake.

As aforementioned, HYDRUS-1D (Simunek et al., 2008) also utilizes the Richards equation (Eq. 3c) with the Mualem-Van Genuchten (Eq. 4) for infiltration. The software was validated on the basis of measured bottom runoff and water level in Meng et al. (2014), which simulated the hydrological performance of two bioretention cells in Beijing, China using almost four years of field data, consisting of water balance and rainfall measurements. The study found that

software was able to accurately predict infiltration and ponding processes, and determined that the hydraulic conductivity of the filter media and surface area size greatly influenced the hydrological effectiveness of the bioretention cell. Performance was evaluated various statistical measure and all results were found to be acceptable with all root-mean-square error, (RMSE) approximating zero, the mean relative error (MRE) close to 0.20, with a coefficient of determination (R^2) higher (>0.9) in the test using artificial rainfall and relatively smaller (>0.6) in the natural rainfall (Meng et al., 2014). HYDRUS-1D was also able to accurately generate data representing the hydraulic conductivity and cumulative fluxes of four experimental bioretention columns in Fakulti Kejuruteraan, Malaysia based on precipitation, runoff, and flow rate data. RMSE was used to validate water outflow and R^2 was used to determine the fit of the model to the soil water storage over time. All RMSE values were approaching zero and R^2 values were above 0.90 (Jitolis et al., 2022).

The review, Lisenbee et al. (2022), found that HYDRUS-1D was the only model evaluated that incorporates plant growth in its calculation of ET. In terms of limitations, it was found that both GIFMod and HYDRUS-1D are incapable of modelling IWS upturned elbow configurations. It was also mentioned that HYDRUS-1D is used in fewer studies due to its high complexity.

The RECHARGE software also uses a Richard's equation to model infiltration (A. R. Dussaillant, 2002). It was used by Wang et al. (2019) for the modelling and field validation of a bioretention basin in Singapore. Normalized RMSE, R^2, and NSE were used validate the model according to water level in the ponding zone (nRMSE = 1.11, R^2 = 0.86, NSE = 0.86), surface overflow culvert discharge (nRMSE = 7.06, R^2 = 0.77, NSE = 0.77), and basin outlet discharge (nRMSE = 2.21, R^2 = 0.30, NSE = 0.21). Despite the relatively poor simulation of outlet discharge (to be considered satisfactory, Moriasi et al. (2015) recommends NSE > 0.5 and R^2 > 0.6), J. Wang et al. (2019) concluded that the RECHARGE model was successful in

representing the performance of the bioretention basin. However, its use of the Richard's equation contributes to its complexity.

DRAINMOD (R. W. Skaggs et al., 2012), another software, utilizes the Green-Ampt process to determine infiltration at the surface as well as between layers (W. A. Lisenbee et al., 2021). The Green-Ampt model (Heber Green & Ampt, 1911, Eq. (5)) is a simplification of the Richards equation based on Darcy's law which has less parameters, reducing input requirements and computation demand (W. A. Lisenbee et al., 2021). However, the Green-Ampt equation requires certain assumptions such as a simplified wetting front movement with one dimensional, vertical flow, where a distinct boundary separates soil with saturated content from soil of constant initial moisture content.

$$i_{eq,GA}(I) = 1 + \frac{(\theta_s - \theta_i)(h_a - h_{wf})}{I} \qquad (5)$$

Where $i_{eq,GA}$ [LL^{-1}] is the equivalent hydraulic gradient, θ_s is the saturated water content, θ_i is the initial water content, h_a [L] is the water pressure head at the surface, h_{wf} [L] is the water pressure head at the wetting front, and I [L] is the cumulative infiltration.

The results of the calibration and validation of DRAINMOD (R. W. Skaggs et al., 2012) by Brown et al. (2013) indicated that the model could be used reliably to predict stormwater hydrology on a long-term basis, drawing attention to the use of the soil-water characteristic curve (SWCC) to account for water in the media. It was inferred that the reliability of the model would increase with more measurements of site-specific parameters as well as media and underlying soil characteristics. Runoff depth, overflow, water table depth were validated using the Nash-Suttcliffe Efficiency (NSE). For both calibration and validation periods, the NSE coefficients for runoff exceeded 0.99, drainage and exfiltration/ET were greater than 0.8, and ranged from 0.6 to 0.9 in a pre-repair phase (Brown et al. 2013). However, a key limitation of

DRAINMOD is its temporal resolution, with daily outputs as the smallest time step produced (W. A. Lisenbee et al., 2021).

DRAINMOD-Urban (W. Lisenbee et al., 2020) is an updated version of DRAINMOD which is able to accept finer resolution inputs and outputs, specialized for representing urban hydrology in bioretention cells. A similar conclusion was drawn in Diab et al. (2022) for DRAINMOD-Urban, where the model showed strong performance for modelling bioretention outflow hydrographs and volumes over a two-year period. This study made use of NSE and PBIAS to evaluate the validity of the model's ability to simulate drainage and overflow (Diab et al., 2022). Although considered to be among the most comprehensive drainage models, DRAINMOD-Urban involves intensive calibration (W. Lisenbee et al., 2020; W. A. Lisenbee et al., 2021)

The latest version of the Storm Water Management Model (SWMM 5) developed by the U.S. Environmental Protection Agency (Rossman, 2010) was also found to be successful in modeling several types of LID controls. It represents surface infiltration using a modified Green-Ampt equation (Chen et al., 2019)Eq. (6)), adjusted to account for ponding depth, and Darcy's law to representing infiltration between layers. A study evaluating SWMM for a catchment in Nyabugoga, Rwanda, showed that it was acceptable for modelling stormwater runoff quantity with calibration and validation values of r^2 above 0.6, NSE above 0.5, and d (index of agreement, (8) above 0.7 (Niyonkuru et al., 2018).

$$f = K_s\left(1 + \frac{(\phi - \theta_0)(d + \psi)}{F}\right) \qquad (6)$$

Where f [LT^{-1}] is the infiltration rate, K_s [LT^{-1}] is the saturated hydraulic conductivity, ϕ [L^3L^{-3}] is the soil porosity, θ_0 [L^3L^{-3}] is the initial soil water content, d [L] is the deoth of ponded

water, ψ [L] is the suction head at the wetting front, and F [L] is the cumulative infiltration volume.

When comparing SWMM and DRAINMOD-Urban, Lisenbee et al. 2022 identified SWMM as the easier-to-use model and better for modelling event volumes in both calibrated and uncalibrated simulations, with NSE ranging from 0.70 to 0.93 for drainage hydrographs and 0.59 to 0.81 for overflow hydrographs. DRAINMOD-Urban performed better for drainage hydrographs when calibrated (NSE =0.6), incorporating a better representation of soil porosity at different depths. Although found to be more sensitive to calibration, all calibrations in DRAINMOD-Urban performed poorly for overflow hydrographs. Both models appeared to struggle with modelling high temporal resolution time series. The results of this study indicated an opportunity to use both models together to optimize their strengths. (W. A. Lisenbee et al., 2022).

SWMM has also been used in conjunction with other models, such as RECARGA (Gao et al., 2018). In this study, SWMM was used to simulate runoff and RECARGA was used to model the hydrological performance of a bioretention cell. It was found that the models together were successful in predicting the runoff reduction rate of a bioretention basin with and without an underdrain.

RECARGA (A. Dussaillant et al., 2003) also uses a modified Green-Ampt method for surface infiltration and Darcy's law for infiltration between layers. A similar model to RECHARGE, it was found to have comparable qualitative and quantitative results for values such as recharge, runoff, and maximum continuous ponding time (A. Dussaillant et al., 2003). The model has limited ability to represent certain IWS configurations and capture peak flows (W. A. Lisenbee et al., 2021).

A one-dimensional model for the simulation of flow through the porous storage of an infiltration system into the surrounding soils was developed by Browne et al. (2008). The saturated zone underneath the storage was represented using Darcy's law, and the unsaturated soils were modelled using a moisture-based solution to Richards equation. The model was found to produce a good fit to observed data, analytical solutions, as well as HYDRUS (Browne et al., 2008). A few years later, Browne et al. (2013) developed a two-dimensional model to simulate flow from porous storage. The authors claimed that a two-dimensional model is better for simulation of stormwater infiltration into trenches as it takes antecedent moisture conditions into account. In this study, unsaturated flow was modelled using a combination of a soil moisture-based Richards solution and Darcy's law. A tracking method was employed to account for saturated flow and a storage model was used for ponding and flow through highly porous storage. The model was confirmed by validation and was found to successfully account for the nonlinear conditions of rapid infiltration into coarse grained soils despite being relatively simple, mathematically. RMSE and NSE were used to measure the performance of the model (Browne et al., 2013).

Daly et al. (2012) presented a stochastic model to describe the statistical behavior of the main variables involved in the water balance of bioretention system. The model was tested against field data from a stormwater biofiltration system in Melbourne, Australia, and emphasized the significance of the variability of inflow regimes when it comes to biofilter performance. The authors expressed a necessity for its introduction into modelling tools, and suggested better consideration of treatment performance (Daly et al., 2012).

The semi-distributed software, CANOE, is more easily configurable and reduces computation time when compared to more complex software such as HYDRUS, although it is often less accurate since it does not take fully into account the variability of physical properties or

variability of precipitations (Ichiba et al., 2013). A multi-outlets model, Hydrobox, was developed by Boutaghane (2016) to be integrated into CANOE as an LID module. Hydrobox was found to simplify construction and calibration of the model. In this study, it was applied to catchment in Yzeron, France, where simulated discharge flows were validated against measured data using NSE and RMSE. The results were encouraging and the platform was found to provide a good representation of flood events (Boutaghane, 2016).

A simplified physically based model for Hydrobox was developed and field-validated by Bonneau et al. (2021) This model attempted to find a compromise between oversimplification without physical basis and physical soundness requiring a large number of parameters. It accounted for ET, overflow, infiltration into the filter, exfiltration to surrounding soils, and underdrain discharge, mainly using Darcy's law and the van Genuchten and Mualem models. The model displayed good performance for simulating outflow rates, although there were discrepancies with the water level in the basin. It also was found to be particularly sensitive to the orifice coefficient of the underdrain pipe. Although ready for incorporation into the LID toolbox of a catchment scale software, the authors suggest further development on how to incorporate more parameters while avoiding overcomplication and more calibration effort. The performance measured used were NSE, RMSE, and PBIAS for outflow and water level (Bonneau et al., 2021a).

3.5 Climate Impacts

Climate change is becoming an increasingly important factor when considering urban drainage systems. The changes in precipitation patterns make cities especially vulnerable to flooding. Based on climate scenarios and perturbed historical rainfall series, derived intensity-duration-frequency (IDF) statistics and design storms showed that storm intensity can increase drastically within the twenty-first century. To account for this, the study found that current

bioretention systems will need a 11% to 51% increase in storage to keep overflow frequency to the current level (Willems, 2013).

M. Wang et al. (2019) examined the effectiveness of urban catchment in response to changes in probable storm characteristics and distributions in Guangzhou, China. Effectiveness was evaluated in terms of hydrological performance in response to different climatic scenarios, resulting in increased peak flows, precipitation frequency, and severity of waterlogging, which occurs when extreme rainfall exceeds urban flood discharge capacity (Duan et al., 2022). The study found that bioretention cells could significantly reduce runoff volume and peak discharge in rainfall events with short return periods, but were less effective in longer-duration storms (M. Wang et al., 2019).

In response to the increased risk of urban flooding due to climate change in Novi-Sad, Serbia, Greksa et al. (2022) analyzed the effect of different bioretention design parameters on flood resilience. The study found that native soil hydraulic conductivity and bioretention size had the greatest influence on runoff reduction. They also commented that ponding time should be less than 24 hours after a storm event has ended, since prolonged water retention time can impact plant survival and hinder hydrological performance of the system (Greksa et al., 2022).

Hathaway et al. (2014) evaluated the hydrological functioning of bioretention systems in North Carolina using data from downscaled climate projections and DRAINMOD. They found that the frequency in magnitude of outflow from the systems would substantially increase, leading to increased amounts of uncontrolled and untreated runoff from contributing wetlands. To account for this, 9 to 31 cm of additional storage would be needed, reflecting the higher importance of surface storage volume and infiltration on system resilience to climate change. The authors recommended more research be conducted in other locations, since the severity of climatic impact varies by region and the impact on rainfall patterns is relatively moderate in

North Carolina (Hathaway et al., 2014). For instance, in Northeast Ohio, where future climate scenarios predicted lower annual rainfall depths, longer dry periods, and hotter temperatures, simulated results of bioretention cell performance showed similar volume reductions to current climate scenarios (Winston et al., 2021).

Since the primary manifestation of climate change is seen as shorter, more intense periods of rain, the effects of the dry periods in between are also worth examining. According to Batalini de Macedo et al. (2019), longer dry periods would have an effect on bioretention performance due to the decrease in soil moisture and increase in water demand of plants impacting infiltration and the overall water balance. In the study, a bioretention cell without an underdrain was monitored during wet and dry periods in São Carlos, Brazil under subtropical conditions. They found that peak flow reduction did not vary significantly between wet and dry periods, however runoff retention efficiency was found to be greater in dry periods. Also, bioretention performance in dry periods was more affected by previous soil moisture conditions and rainfall rate, whereas wet periods were impacted more significantly by total precipitation and maximum rainfall intensity (Batalini de Macedo et al., 2019)

Their suggested improvements for bioretention modelling included the consideration of vegetation and plant water use, temporal variation of soil properties, as well as the need for more calibration and validation studies.

3.6 Evaluating Resilience

Folke et al. (2010) defines resilience as “the capacity of a system to absorb disturbance and reorganize while undergoing change so as to still retain essentially the same function, structure and feedbacks coping with uncertainty.” In the same article, the author explains that the concept of resilience is therefore also intertwined with adaptability, since it is necessary for a system to adapt to new conditions in order to maintain its function.

There are many methods for assessing the resilience of a stormwater management system. Dessai et al. (2007) defines two main schools of thought: “top-down” and “bottom-up” reasoning. The more common, prediction-based, “top-down” approach considers the changing climate and then uses it to quantify the expected impact on the system of interest. The “bottom up” approach begins with the impacted system and then assesses its performance in response to different scenarios. Gersonius et al. (2012) points out that problems with the “top-down” method can arise from lack of precision in climate modelling and resulting uncertainties present in climate change predictions.

Kwadijk et al. (2010) describes a “bottom-up” resilience assessment method that utilizes adaptation tipping points (ATP), points where a system exceeds an acceptable threshold. In the case of bioretention resilience, the tipping point would be the magnitude of climate change at which the bioretention system’s capacities are exceeded. This approach is said to focus “on defining if and when adaptation strategies are needed” (Kwadijk et al., 2010). The study found that the method was successful and easy to employ when used to assess the Netherlands water management system’s vulnerability to the effects of climate change.

In order to address the criticism of the “reactive attitude” associated with the ATP method, Gersonius et al. (2012) developed the “Mainstreaming Method” which involves the analysis of ATPs as well as adaptation opportunities in the stormwater system. Although the study found that ATPs are very useful for quantifying the resilience of existing systems, the Mainstreamng Method adds another dimension of adaptation that can be integrated into a system at an earlier stage. Figure 7 shows the steps of the ATP resilience assessment, based on Kwadijk et al., (2010), with the addition of the mainstreaming method’s adaptive potential assessment.

Panos et al. (2021) employed the mainstreaming method to assess the resilience of urban drainage systems in Denver, CO, to climate change. Using this approach, the resilience

assessment indicated at what changes in rainfall regime (tipping points) the neighbourhood would exceed the acceptable conditions. In this case, the threshold values applied (correlating to step two in Figure 7) were calculated based on the urban drainage and flood control standards for the City of Denver. As a result, an investigation into the use of stormwater control for improving tipping points and increasing system resilience was carried out.

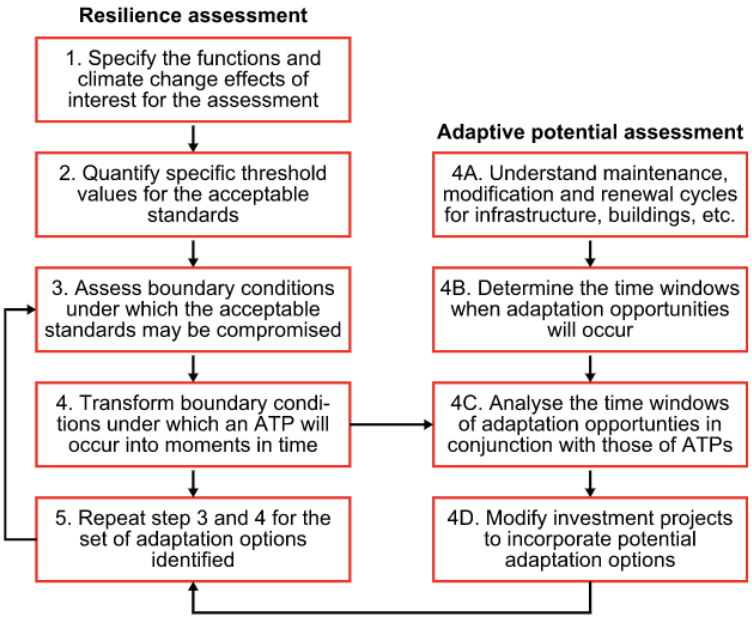

Figure 7. Flow chart of the Mainstreaming method, based off the ATP resilience method, titled "Resilience Assessment," (Gersonius et al., 2012).

3.7 Conclusion

Bioretention systems must be able to operate effectively in the context of climate change, and models must have the capacity to represent their hydrological function accurately. There is a demand for more scientifically and physically based for the design and assessment of bioretention systems (Adu Boateng et al., 2023; Al Sayah et al., 2022), which models should be able to account for (Ichiba et al., 2015; Van Seters & Graham, 2014; Vijayaraghavan et al., 2021). A shortness of long-term bioretention studies was also identified, which are especially

helpful for the validation and calibration models – another area of research that is lacking (Van Seters & Graham, 2014; Vijayaraghavan et al., 2021). For the models that have been validated by long-term studies, they often make use of monitoring data from bioretention cells in a single location, therefore not measuring the model's flexibility for different climates (Boutaghane, 2016; Daly et al., 2012; Jitolis et al., 2022; Meng et al., 2014; J. Wang et al., 2019). Especially since different locations will be impacted by climate in different ways (Hathaway et al., 2014; Rosenberg et al., 2010; Willems, 2013; Winston et al., 2021), it is important to diversify monitoring data to account for models' ability to assess bioretention resiliency.

Present-day hydrological models struggle to find a balance between accuracy and simplicity. Models for low impact development are often either highly complex and intensive in terms of calibration and processing time, or simpler at the cost of accuracy. Studies attempting simplification have often concluded that results could be improved with more parameters, requiring large amounts of specific data (Bonneau et al., 2021a; Boutaghane & Ouerdachi, 2012). Therefore, there is a need for models that are able to achieve high precision while maintaining ease of operation.

To address this, three physically based infiltration models were developed by Asry et al., 2023 as part of the Canoe Hydrobox (CH) platform. Based on approaches similar to the Green-Ampt model with the incorporation of the equivalent hydraulic gradient and the equivalent hydraulic conductivity to determine the water flux, these models attempt to achieve a level of accuracy similar to that of Richard's equation with lower calibration effort.

4 RESULTS AND DISCUSSION

4.1 Literature Review and Data Acquisition

The authors of Batalini de Macedo et al. (2019) and J. Wang et al. (2019) were able to provide monitoring data at 1-minute timesteps for the bioretention cells in Brazil and Singapore,

respectively. A data bank was compiled including all of the data acquired, as well as that of the studies with relatively low temporal resolution.

A lack of diverse field hydrological monitoring studies was identified. Most studies were found to be conducted in the Eastern United States and large urban areas of Australia, with little representation of other climates. Additionally, it appears that there are relatively less long-term monitoring studies. This observation is corroborated by findings of the reviews of Spraakman et al. (2020) and Liu et al. (2014). This could be attributed to cost – funds associated with high temporal resolution equipment and long-term establishment for monitoring in a field setting exceed those associated with small-scale, short-term pilot studies.

4.2 Description of case studies

In order to use models to assess the reliance of bioretention systems, data with varying rainfall intensities and from bioretention cells with different configurations in various climates must be used to gauge the versatility of said models to represent bioretention performance in different climate scenarios. Four case studies from different climates and with varying configurations were identified.

4.2.1 São Carlos, Brazil

Batalini de Macedo et al. (2019) a bioretention cell located in Campus 2 of the University of São Paulo/São Carlos in Brazil. According to the Köppen climate classification system, the site has a humid subtropical climate (Cfa) which features high temperatures and evenly distributed precipitation during the year (Arnfield, 2023). Annual precipitation averages 1404 mm per year and the average temperature is 21°C (Climate-Data.org, 2023). The rainy, or “wet,” season takes occurs from November to April with higher temperatures, while the dry season takes

place from May to October with relatively lower temperatures. The months with the most and least cumulative rainfall are January (274.7 mm) and July (28.3 mm), respectively.

The bioretention basin configuration follows a pipeless design (Figure 2) with a surface area of 61 m^2 and a depth of 3.5 meters, capturing runoff from an area of about 2.3 ha. It features two distinct layers: a 0.3 meter ponding zone for surface storage followed by a 3.2 meter filter layer which acts as subsurface storage. The filter layer itself is comprised of three layers: 0.5 meter topsoil composed of natural local soils with a hydraulic conductivity of 2.0 x 10^ 2 cm/s featuring two types of plants, 0.7 meter intermediate gravel layer with a porosity of 0.4, and a 2 m bottom layer of coarse sand with a porosity of 0.3. The bottom of the basin is 3.5 meters above the water table. See Figure 8 for visualization of the site and the bioretention cell configuration.

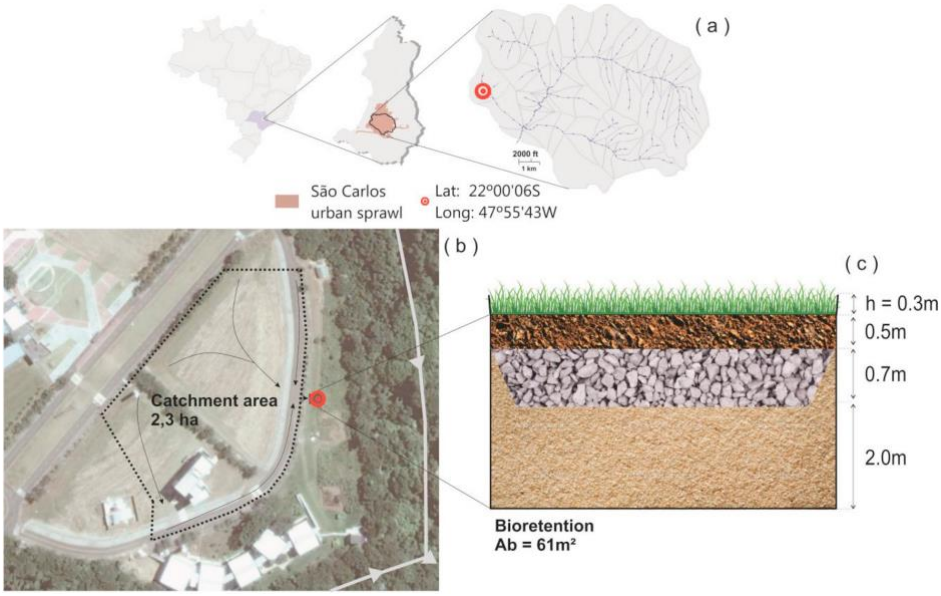

Figure 8. “Bioretention LID practice study area. (a) Its location in the urban basin of Sao Carlos, SP, Brazil; (b) the contribution area in detail; and (c) the internal composition of the LID bioretention practice” (Batalini de Macedo et al., 2019).

29 rainfall events were monitored from 2016 to 2018. The campaign was focused on representing each season, recording events from both the dry and wet season. Precipitation was measured using an on-site rain gauge, although the type was not specified. The water balance variables and data acquisition in the field include: inflow, surface and subsurface retention, overflow, precipitation, percolation, and ET. Runoff inflow and overflow were measured with using water level sensors mounted onto weirs – a compound weir was used for inflow and a triangular weir for overflow). Water level sensors also were used to measure surface and subsurface storage, as well as a piezometer specifically for the water level in the filter media. The mass balance equation was used to estimate percolation, and ET was determined using the Penman-Monteith model (Walter et al., 2005). 1-minute data resolution was applied to all variables. For a more detailed account of the design and monitoring of the bioretention cell, see Batalini de Macedo et al. (2019).

Batalini de Macedo et al. (2019) were able to provide the water balance data of 29 events at 1-minute timesteps with rainfall depths between 2.2 mm and 38.4 mm. Measurements included rainfall depth, inflow and outflow rates, and surface and subsurface storage volumes.

4.2.2 Singapore

The Balam Estate Rain Garden bioretention basin in Singapore was monitored by (J. Wang et al., 2016) The climate of Singapore is classified as a wet equatorial climate (Af) features consistently high temperatures and intense precipitation (Arnfield, 2023). The city-state is characterized by two monsoon seasons separated by intermonsoonal seasons. On average, temperature is 26.7°C and precipitation is about 2366 mm per year (Climate-Data.org, 2023). Although it does not have distinct wet or dry seasons, higher rainfall typically extends from November to January, with February being the driest month (*Climate of Singapore,* n.d.).

With a longer-lasting submerged zone Figure 4, the basin has a surface area of 240 m^2 and receives runoff from an area of 16,800 m^2. It consists of four distinct layers beneath the ponding zone of 13.7 cm: filter layer composed of sandy-loam followed by a sand transition layer, a hard rock and wood chip saturated zone, and a fine gravel drainage at the bottom. The filter media is underlain by an HDPE liner, preventing exfiltration into surrounding soil. J. Wang et al. (2019) summarized the hydraulic parameters of each layer in Table 2. Water can enter the basin through a single inlet and exit through a culvert for surface overflow and an outlet for subsurface drainage. The outlet is 35 cm above the underdrains, allowing for a 35 cm saturated zone to be maintained. See Figure 9 for a schematic cross section of the basin.

Table 2. "Hydraulic parameters of bioretention basin soil media" (J. Wang et al., 2019)

	Unsaturated layer	Transition layer	Saturated layer	Drainage layer
Soil	Sandy loam	Fine sand	Hard rock & wood chips	Fine gravel
Thickness (cm)	40	10	30	5
θ_s (cm^3/cm^3)	0.3	0.35	0.4	0.4
θ_r (cm^3/cm^3)	0.12	0.1	0.4	0.4
K_s (cm/hr)	1	2	100	200
α (cm^{-1})	0.075	0.075	0.075	0.075
n (−)	1.89	1.89	1.89	1.89
m (−)	0.47	0.47	0.47	0.47

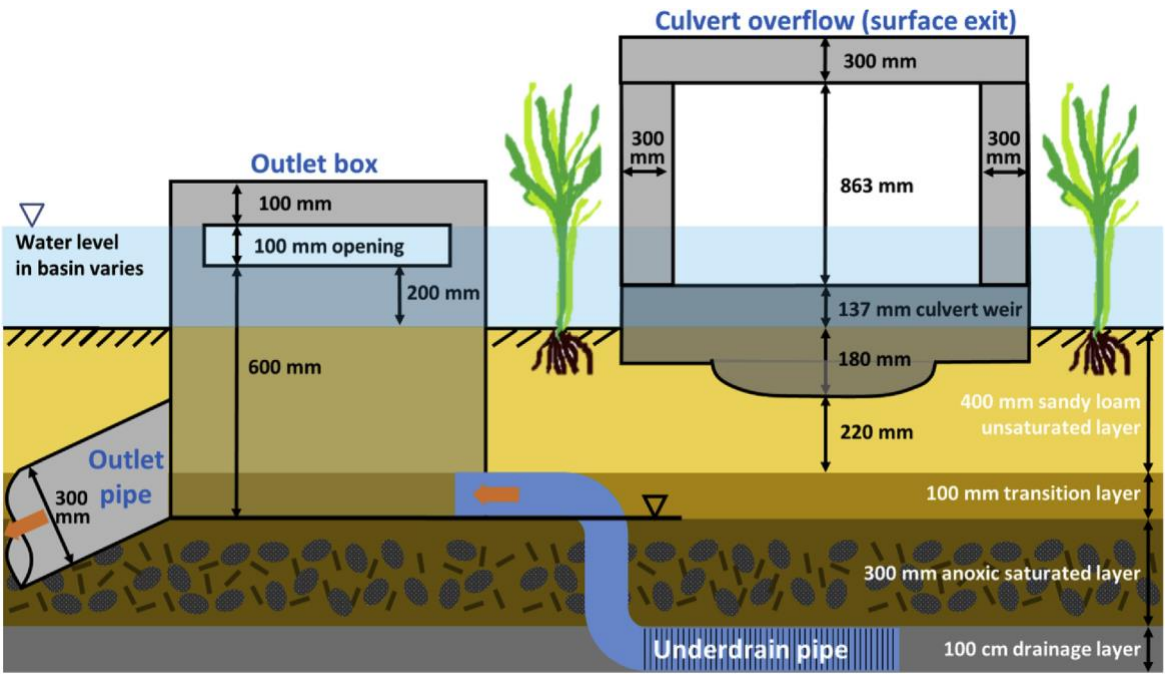

Figure 9. "Schematic cross section of the Balam Estate bioretention basin" (J. Wang et al., 2019)

The monitoring campaign took place from June 2013 to November 2013. Continuous flow rates at the inlet, outlet and culvert, as well as ponding water level at 1-minute intervals were monitored using ultrasonic area-velocity level sensors. A tipping bucket rain gauge was used to measure rainfall at 1-minute timesteps. A plan view of the basin as well as the instrumentation used on site can be seen in Figure 10.

J. Wang et al. (2019) were able to provide water balance data for seven events with rainfall depths ranging from 9.8 mm to 41.4 mm. The data consisted of precipitation depth, inflow, outflow through the outlet pipe, overflow, and water level in the ponding zone.

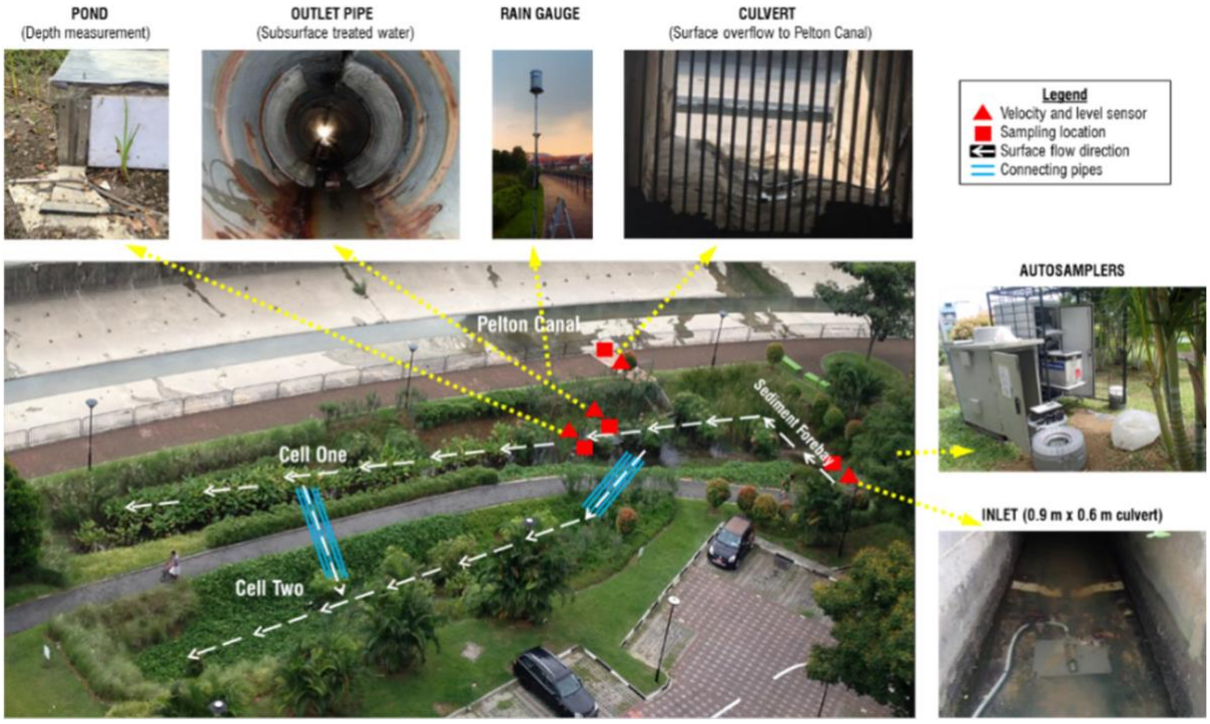

Figure 10. "Plan view of Balam Estate Rain Garden showing the on-site instrumentation" (J. Wang et al., 2016)

4.2.3 Omaha, Nebraska, USA

Two bioretention gardens with temporary submerged zones (Figure 3) about 1.5 km apart were monitored by Strauch et al. (2016) Omaha Nebraska. The Köppen classification for Omaha is a hot-summer humid continental climate (Dfa), characterized by contrast in temperature across seasons (Arnfield, 2023). However, it is relatively temperate with significant rainfall throughout the year, averaging 11°C and 926 mm of rainfall annually (Climate-Data.org, 2023). Omaha sees drier summers with July (27mm) as the driest month, and wetter winters with January (264mm) as the rainiest month.

Both gardens have similar general composition in terms of filter media, consisting of soils with firm to stiff clay fill material above a firm, low plasticity clay. They both feature underdrains wrapped in coarse gravel and a geotextile fabric filter, which rest in an infiltration trench

overlayed with a fine sand and compost mixture. The topsoil layer is 6 in (0.15 m) in depth and consists of a combination of compost and natural soil.

One of the bioretention gardens is located in the Douglas County Health Center (DCHC) and consists of two gardens joined in series. The first garden has a surface area of 2,100 ft^2 (195 m^2) and ponding zone 0.39 ft (0.12 m) deep; when the ponding depth exceeded, the overflow enters the second garden through a channel. Overflow in the second garden flows into a combined sewer. Due to errors in the experiment, (Strauch et al., 2016) excluded the second garden in the series from all computations. See Figure 12 for a plan view of the DCHC bioretention site and **Error! Reference source not found.** for a schematic diagram of the garden's configuration.

Figure 12. "Bioretention garden at the Douglas County Health Center site, Omaha, Nebraska. A, drainage area and bioretention garden 1 layout" (Strauch et al., 2016)

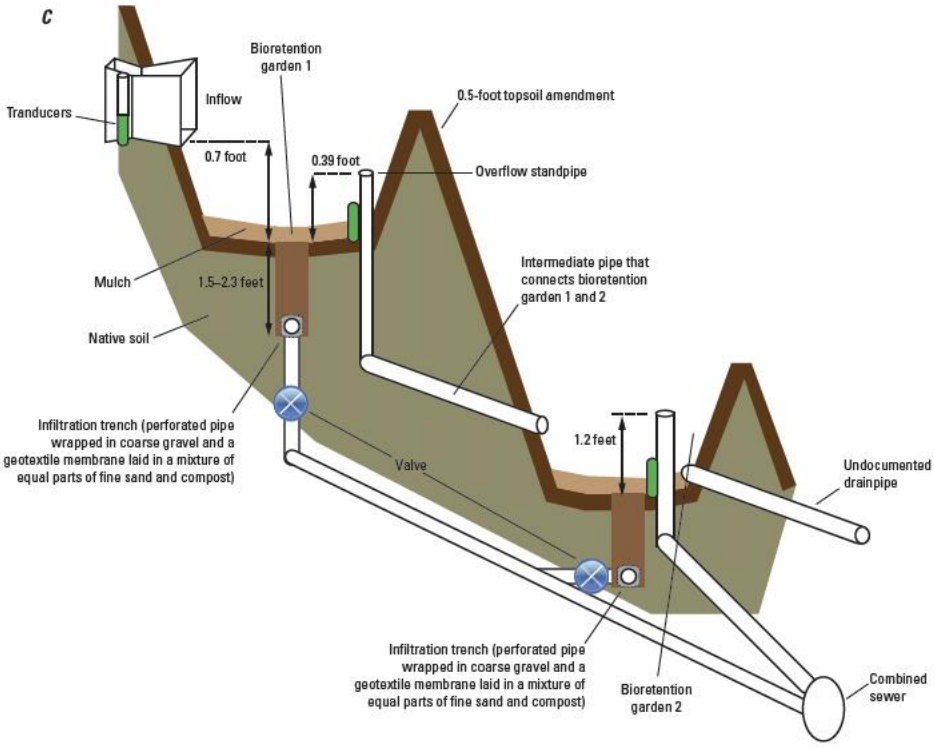

Figure 11. Cross sectional diagram of the DCHC bioretention garden (Strauch et al., 2016)

The other bioretention cell was located at the Eastern Nebraska Office on Aging (ENOA) and has a surface area of 2,900 ft^2 (270 m^2) as well as a ponding depth of 1.66 ft (0.51 m). See Figure 13 for a plan of the site and for a schematic cross section of the garden.

Figure 13. "Bioretention garden at the Eastern Nebraska Office on Aging site, Omaha, Nebraska. A, drainage area and bioretention garden layout" (Strauch et al., 2016)

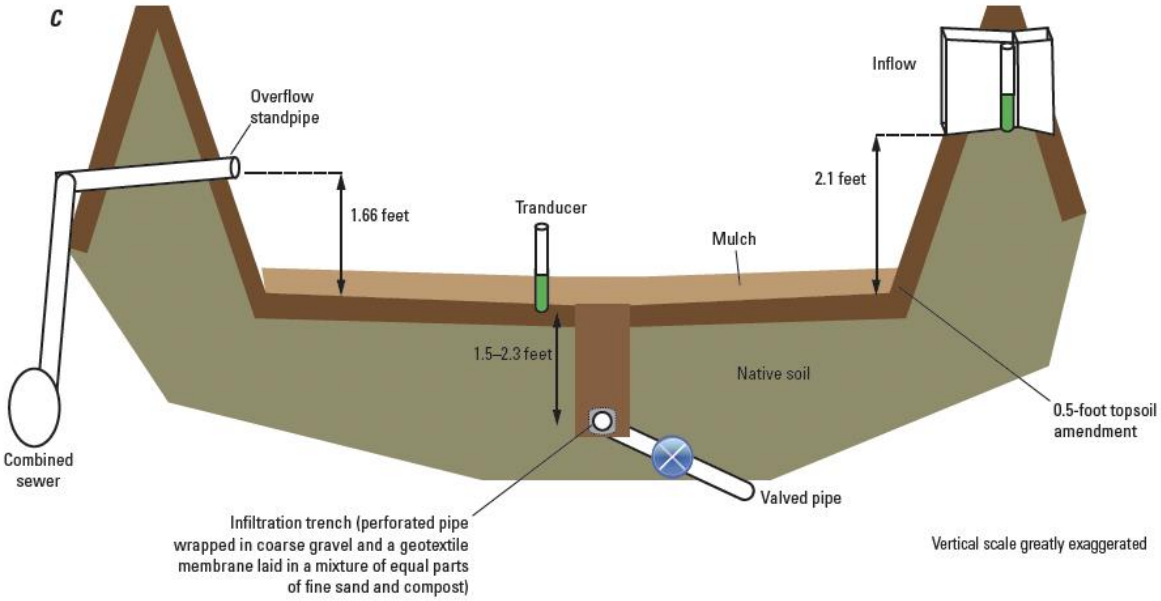

Figure 14. Schematic cross section of the ENOA bioretention garden (Strauch et al., 2016)

The monitoring campaign took place over the course of four years, from May through October in 2011 and from April through October in 2012, 2013, and 2014. Precipitation was measured in 15-minute intervals suing a tipping bucket rain gauge. A Parshall flume with a 9-in (0.23-m) contraction and an attached stilling well was used to measure flow into the gardens with pressure transducers to account for water level in the stilling well. Pressure transducers were also installed at the base of the gardens to measure water depth related to overflow. Evapotranspiration was estimated using the Penman-Monteith equation, and other water balance values were calculated using the water balance equations.

Although a response from the authors from this study was received, the temporal resolution was too low to be used for CANOE Hydrobox. The only available data consisted of event summaries with the total rainfall depth, inflow and outflow volumes, subsurface storage as well as peak inflow and outflow rates for each event. See Strauch et al. (2016) for a more comprehensive overview of the site and data.

4.2.4 Vaughan, Canada

Van Seters & Graham (2014) monitored a bioretention cell in the city of Vaughan, near Toronto, Canada. The climate of Vaughan is characterized as cold and temperate with significant precipitation, falling under the Köppen classification of warm-summer humid continental (Dfb, Arnfield, 2023). The average temperature is 8.3°C with about 845 mm of rain per year. Despite consistent rain, the period between April and December tends to be wetter (Climate-Data.org, 2023).

The bioretention cell area consists of basic design (Figure 5) a 123 m^2 linear island in a parking lot with a 128 m^2 bump out on each end, capturing runoff from an area of 2,272 m^2 Figure 15. The system is also connected to an 84 m^2 swale via underdrain 1.3 m below the surface. Filter media consisting primarily of sand was placed in the center island portion of the cell, while the swale section to the East used finer textured native soils. In both cases, the media was underlain by a geotextile filter and a 65 cm stone storage above a perforated underdrain (Figure 16).

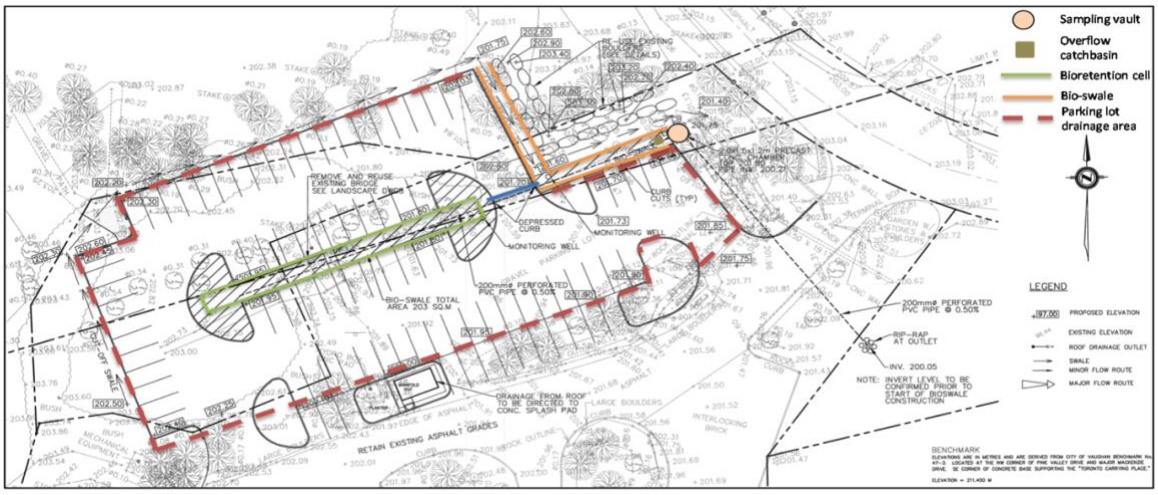

Figure 15. "Study site showing [Vaughan] parking lot drainage area, bioretention cell, bioswale, overflow catch basin and sampling vault" (Van Seters & Graham, 2014)

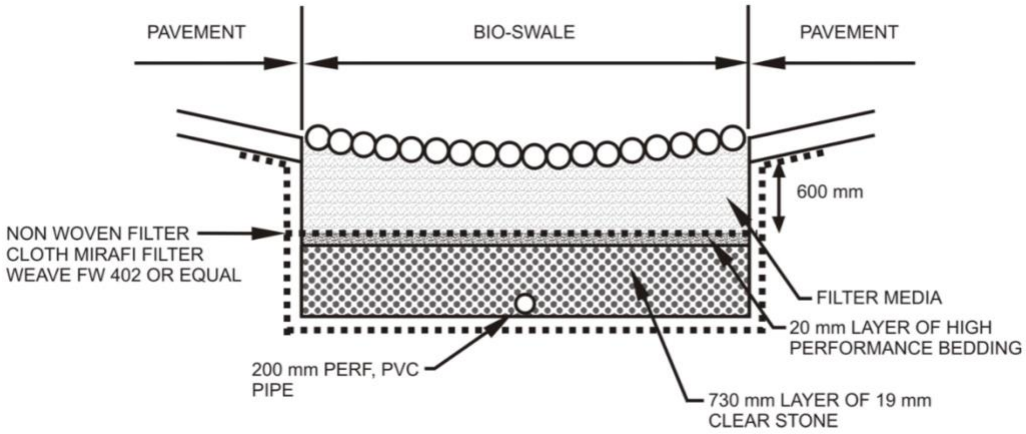

Figure 16. "Cross section of the [Vaughan] bioretention cell. The cell surface includes a combination of bioretention plants and river stone. Bump-outs are fully vegetated." (Van Seters & Graham, 2014)

The cell was monitored year-round from January 2011 to November 2012. A tipping bucket rain gauge was used on site to measure precipitation, and meteorological station 2 km away was used as a secondary source of rain data. Surface overflows and underdrainage were monitored using a tipping bucket flow meter and an automated sampler in a sampling vault. Water level in the ponding zone was measured at a 10-minute temporal resolution with a pressure transducer. Infiltration measurements were also conducted on site using a Guelph permeameter. The following water balance variables were tabulated: rainfall duration, depth, max rainfall intensity, runoff volume, maximum water level and duration above and below the surface, as well as underdrain flow volume, estimated overflow volume, and runoff reduction.

Similar to the Omaha site, the data available for this study included only event summaries and therefore could not be used to validate the CANOE Hydrobox models.

For more information on the bioretention cell design or monitoring campaign, consult (Van Seters & Graham, 2014).

Figure 17. Monitoring site for the Vaughan bioretention cell (Van Seters & Graham, 2014)

4.3 Case study performance

When comparing the the Singapore and Brazil bioretention cells under similar rainfall regimes in Table 1, a few patterns were identified that could be attributed to the configurations of the respective cells. See the Supplementary Materials section (p.56) for all associated figures.

In terms of overflow, the Brazil system generally experienced relatively lower volumes of overflow that lasted shorter amounts of time. This could be attributed to its lack of liner,

therefore facilitating overall infiltration since there was nothing preventing exfiltration into surrounding soils. The Singapore cell had an impermeable liner which prevented all exfiltration, which could contribute to higher volumes of overflow for the same reason. Additionally, the only outflow other than overflow in the Singapore system was an outlet pipe.

System behavior in terms of surface storage, or the volume of water kept in the ponding zone of the bioretention cell, also varied. The surface storage volume in the Singapore system was significantly higher in each event due to the larger surface area, and therefore storage capacity, of the bioretention cell. Additionally, water stayed in the ponding zone for longer which can likely be attributed to the same reason.

Subsurface storage, or the volume of water kept in the filter, was not compared since that data was not made available for the Singapore case study. Water level is another parameter that can be compared, and can easily be calculated by dividing the storage volume by the surface area of the cell. Through this comparison some components of bioretention system design that have an effect on function were identified. This includes the surface area of the system, piping, and liners.

4.3 Validation

The models CH1-3 were not able to be incorporated into the CANOE software within the timeline of this project. Further technical difficulties with the access key restricted all use of CANOE, earlier versions included.

However, the code in R used for the model CH 1 developed by Asry et al. (2023) was used. Due to time constraints, only seven events from the São Carlos, Brazil bioretention monitoring campaign were used for validation, and calibration was not possible. See Figure 18 for the results of the hydrograph simulation compared with the observed data.

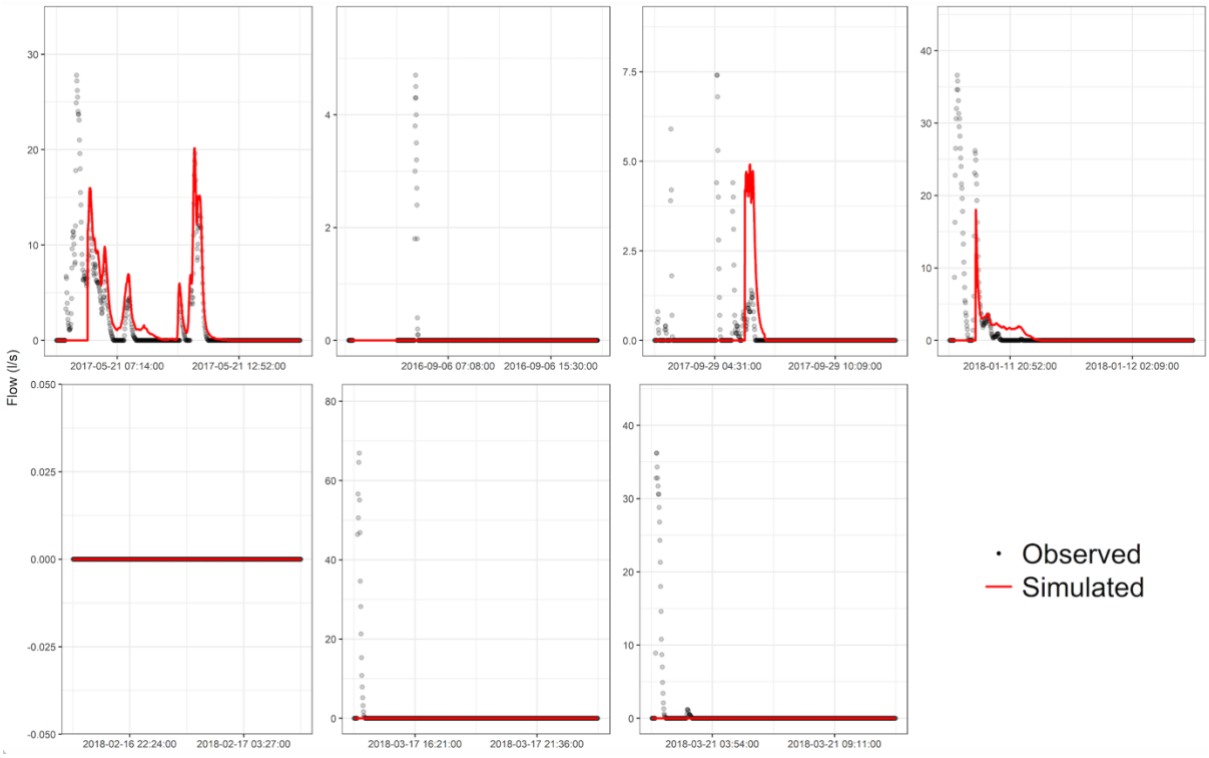

Figure 18. Simulated and observed hydrographs for seven events monitored in the São Carlos, Brazil, bioretention cell.

Performance measures (NSE, PBIAS, RMSE) were applied to these results, which showed that the model could not be validated. As shown in Figure 19., the NSE is most concentrated at the value of zero, which indicates that the mean of the data has relatively the same predictive ability as the model (Moriasi et al., 2015). Furthermore, the PBIAS measurement was completely unsuccessful and the RMSE was unevenly distributed. The unsatisfactory performance of the model is likely due to the fact that it was not properly calibrated.

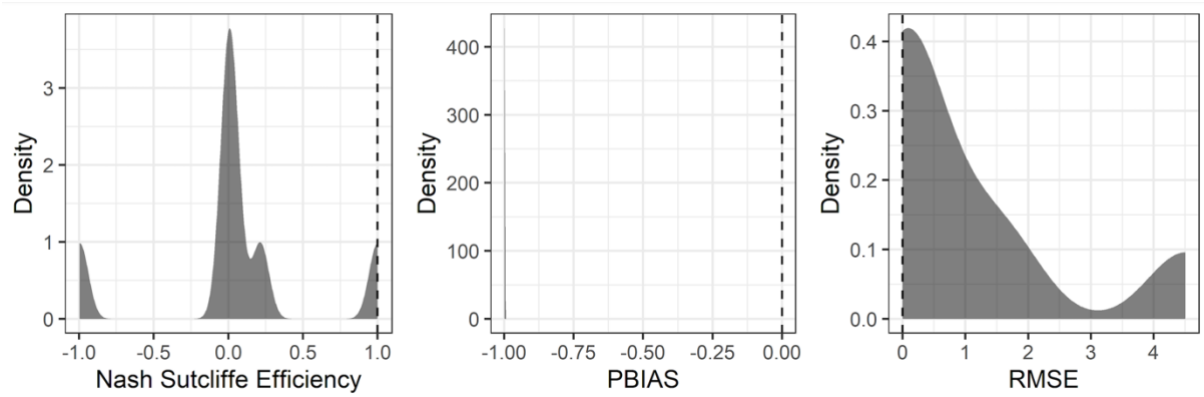

Figure 19. Performance measures (NSE, PBIAS, RMSE) for the ability of CH 1 to accurately simulate the observed hydrographs for the seven events monitored in the São Carlos, Brazil, bioretention cell.

4.4 Resilience Assessment

A method for assessing the resilience of bioretention systems was developed incorporating an ATP process adapted from the mainstreaming method (Gersonius et al., 2012). A visual representation of the method developed for the purpose of this study can be seen in Figure 20. The first step calls for the definition of a performance baseline of the bioretention system. This encompasses the behavior of the bioretention system under regular conditions, based on monitoring records. When making an assessment in the context of increased rainfall and flood risk, performance can be evaluated on the basis of factors such as runoff reduction and overflow frequency. In contexts of dryness, evapotranspiration can be considered.

Next, the evaluator is to choose a climate change effect of interest. For instance, this could be increased precipitation. Based on this effect, a point of failure is to be identified. This can be done by referring to past monitoring data to identify the point at which the system fails relative to the defined base performance standards. Using the effect of increased precipitation, this point of failure could be the runoff volume at which a bioretention basin is no longer able to uphold said base performance.

The point of failure identified can then be used to calculate a specific climate change effect. Following the example, the runoff volume found in the previous step can be divided by the surface area of the bioretention cell to find the total rainfall depth of a storm that would hypothetically cause the "failure" of the system. Then, using a calibrated and validated hydrological model, the behavior of the bioretention cell of interest would be observed under a simulation of a storm with the characteristics calculated in the previous step. Based on the simulation, the performance of the bioretention cell can be evaluated in regards to predefined regulations, such as recovery time within 24 and 48 hours (Maine Stormwater Best Management Practices Manual, 2016).

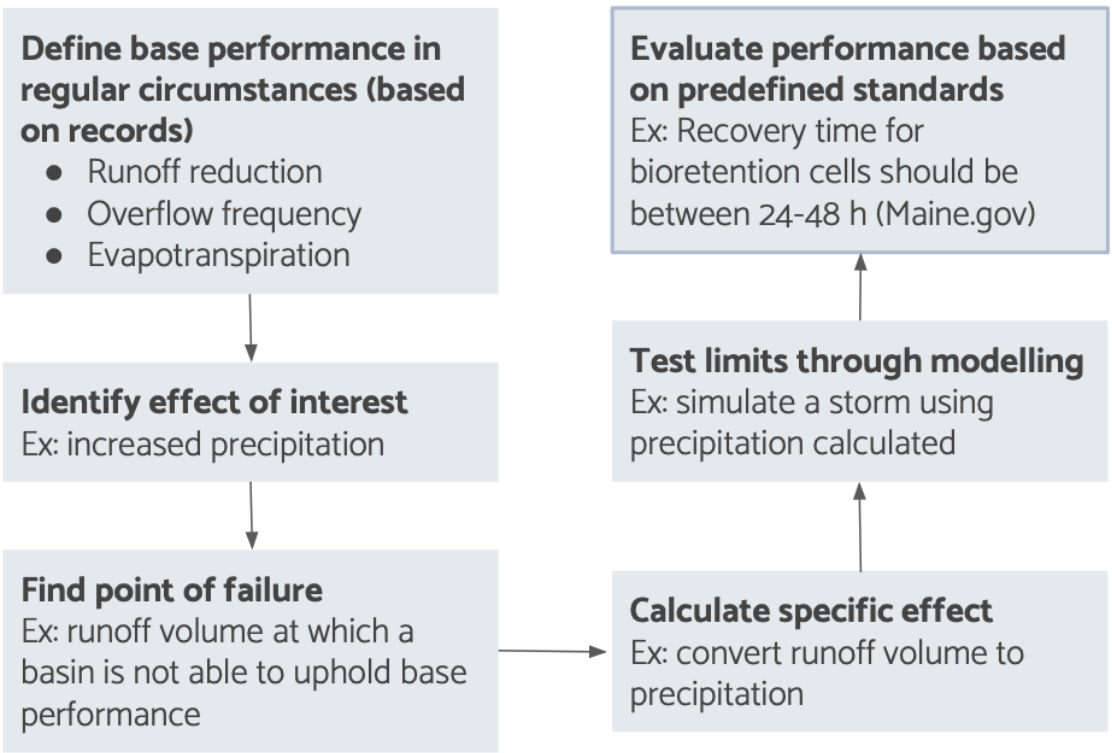

Figure 20. Method for assessing the resilience of bioretention systems using an ATP process

5 CONCLUSION

The objectives of this study were to present a review of previous research in the domain of bioretention monitoring and modelling, to cultivate field monitoring studies of bioretention

systems in varying climates and configurations, and to develop a method for testing the resilience of said systems.

Four long-term bioretention monitoring studies were found for systems with various configurations located in different climates (Köppen Cfa, Af, Dfa, and Dfb). Authors were contacted for data with high temporal resolution. However, only data from the Brazil and Singapore bioretention systems were able to meet the standards of the models to be validated (Asry et al., 2023). The performance of the two systems were compared under similar rainfall regimes, and factors affecting function were identified (surface area, piping, liners). Additionally, a method for evaluating the resilience of bioretention cells incorporating climate change effects and the use of hydrological models was adapted.

Due to technical difficulties with the CANOE software, full validation and calibration of the models CH1-3 was not successful and resilience assessments could not be executed. However, a data bank was compiled with all of the monitoring data acquired. This data is to be used to validate the CH1-3 models once the CANOE software is working again. The data deemed unsuitable was also included in the data bank in the event that CANOE is updated to gain the ability to use lower-resolution data.

In terms of modelling, there is a need for physically-based models that are able to accurately simulate bioretention infiltration without being too complex or difficult to calibrate. It is also important that these models be flexible. The models CH1-3, were found to be lacking in this regard, since their associated code in R was not able to be adapted to the selected bioretention systems in time for this thesis. Additionally, more field validation of bioretention models is necessary, especially in different climates and varying weather regimes – there is a lack of geographic diversity and high temporal resolution in hydrological monitoring of bioretention

cells in field settings. In terms of climate adaptation, resilience assessments with standardized procedure are necessary to evaluate bioretention systems' durability and needs for amendment.

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7 SUPPLEMENTAL MATERIALS

7.1 Singapore

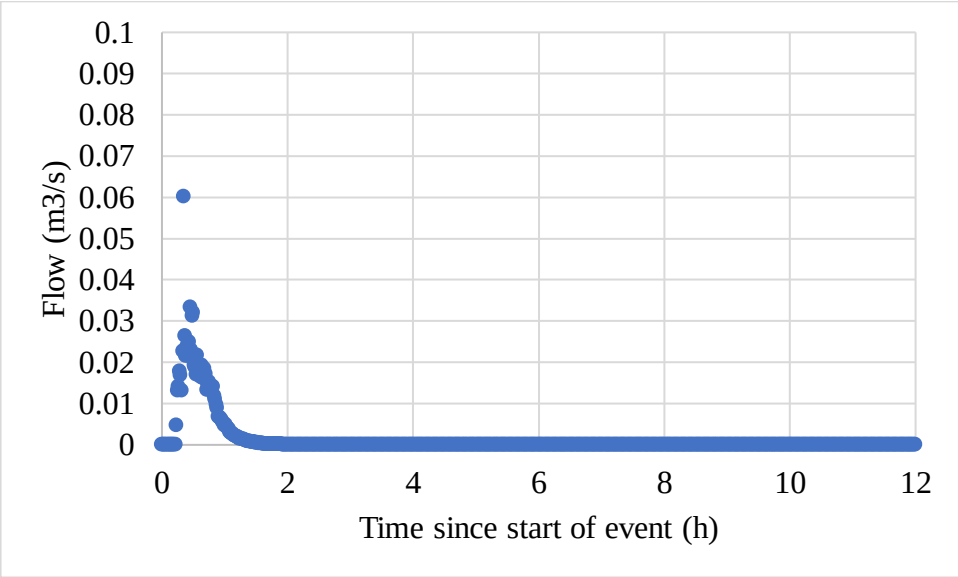

Figure 21. Overflow over time for a storm with total rainfall depth of 9.8mm

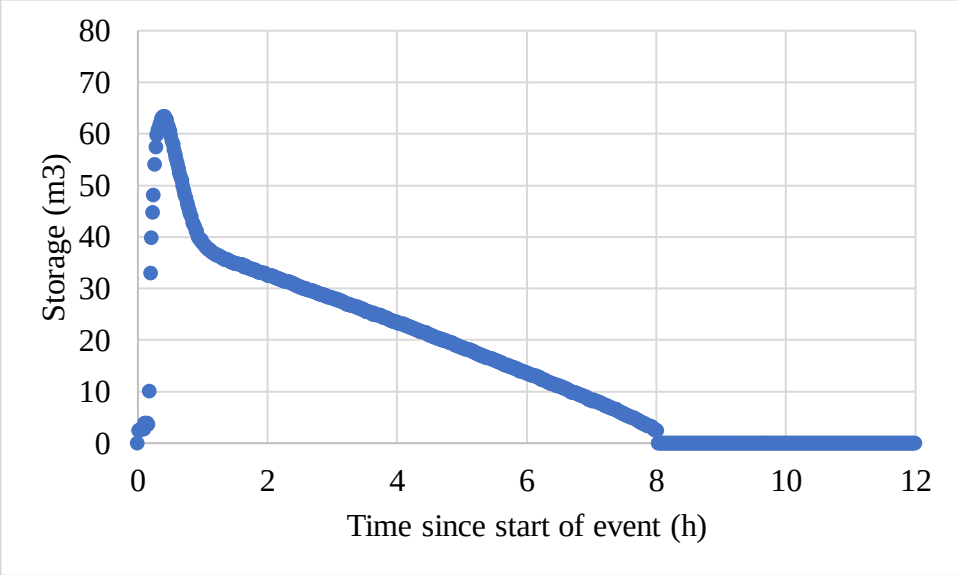

Figure 22. Surface storage over time for a storm with total rainfall depth of 9.8mm

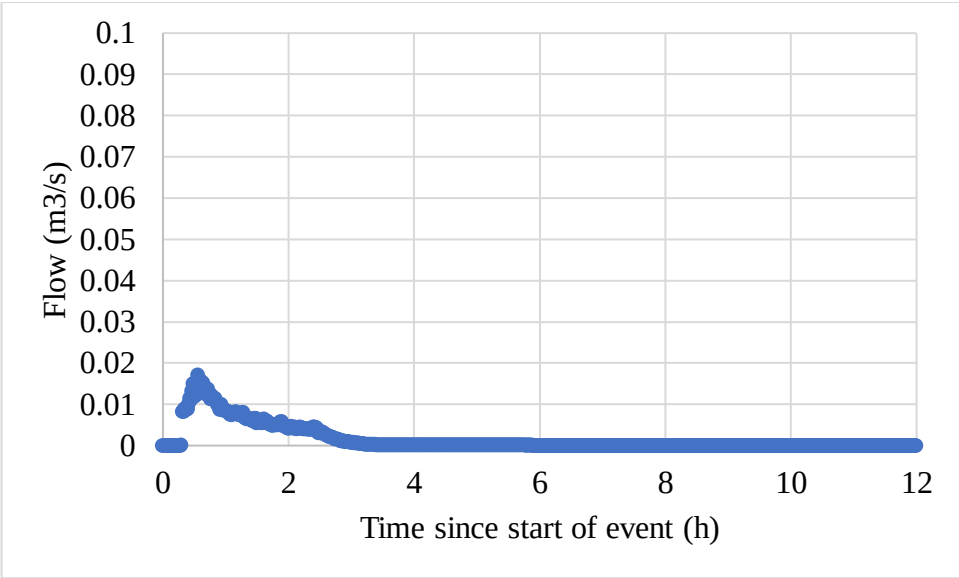

Figure 23. Overflow over time for a storm with total rainfall depth of 11.6mm

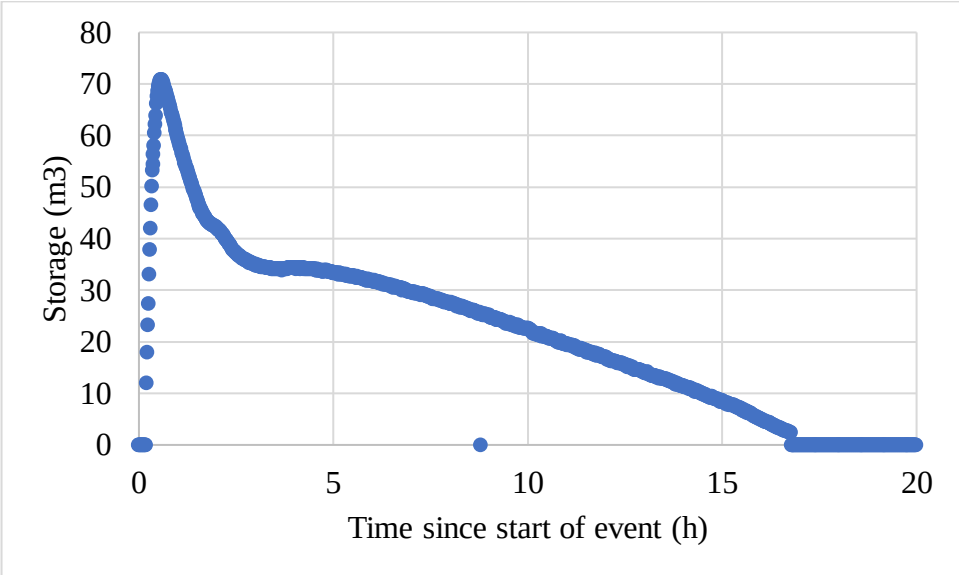

Figure 24. Surface storage over time for a storm with total rainfall depth of 11.6mm

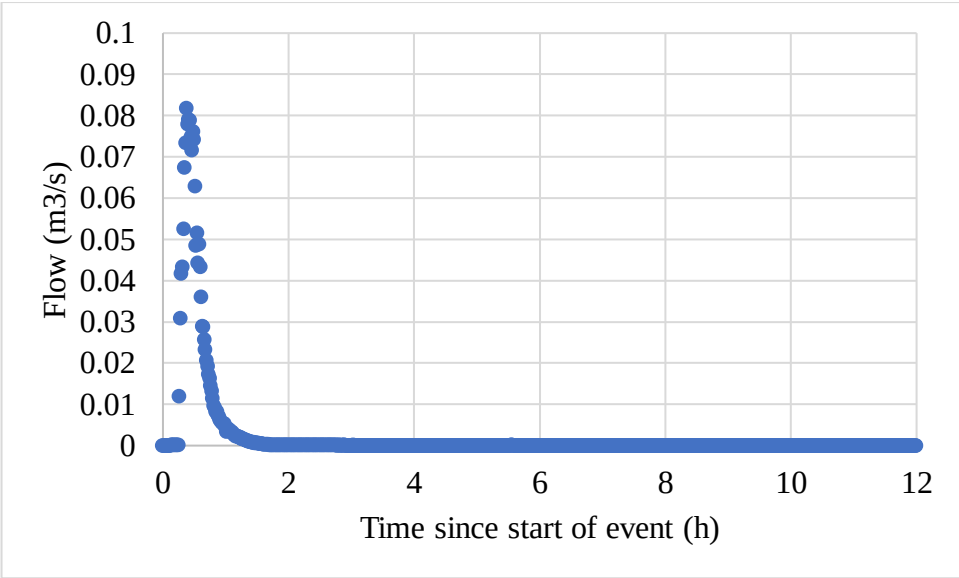

Figure 25. Overflow over time for a storm with total rainfall depth of 11mm

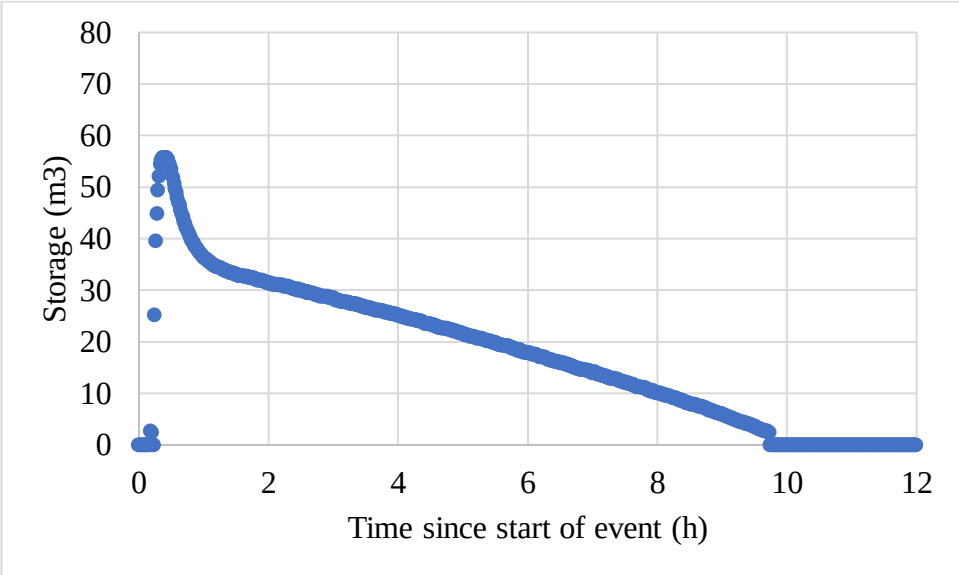

Figure 26. Surface storage over time for a storm with total rainfall depth of 11mm

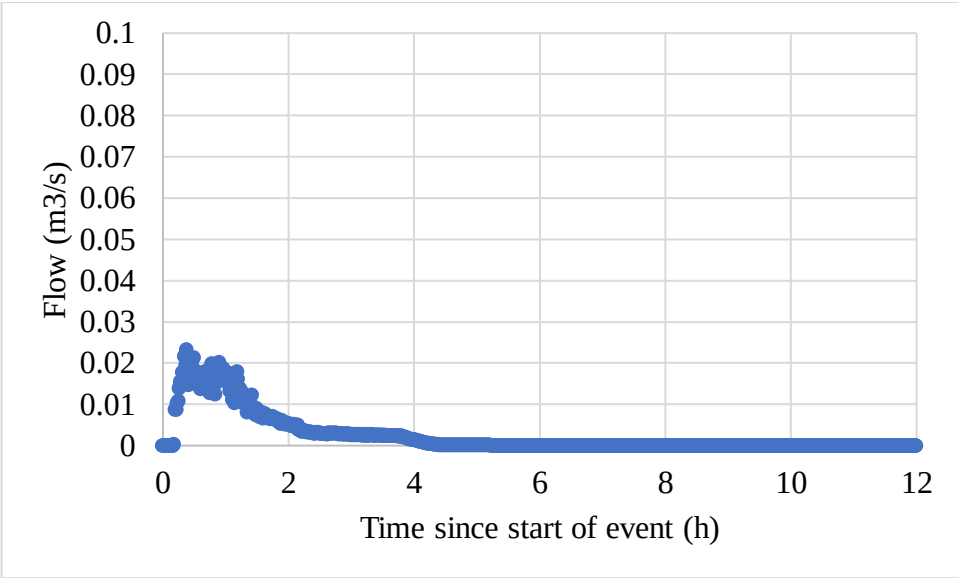

Figure 27. Overflow over time for a storm with total rainfall depth of 18.8mm

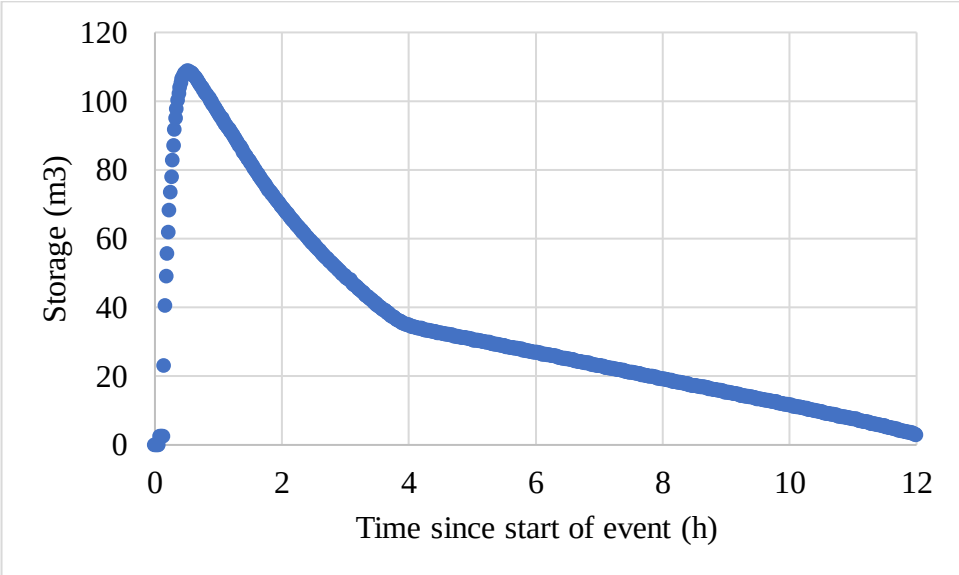

Figure 28. Surface storage over time for a storm with total rainfall depth of 18.8mm

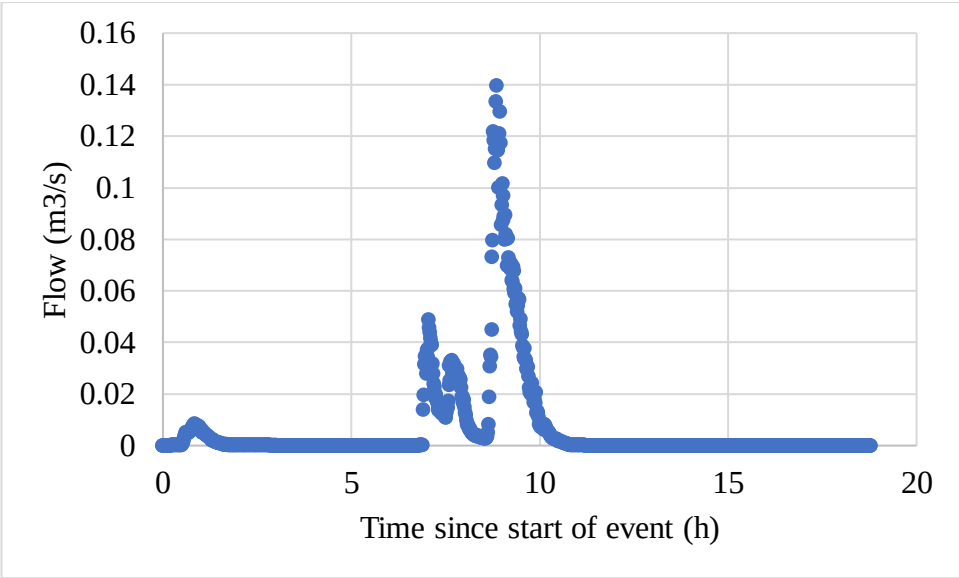

Figure 29. Overflow over time for a storm with total rainfall depth of 41.4 mm

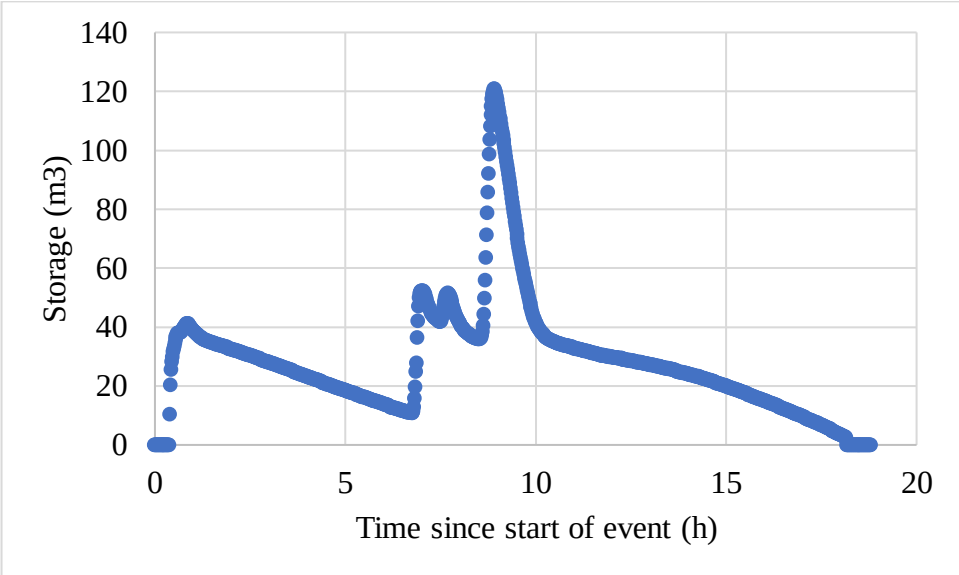

Figure 30. Surface storage over time for a storm with total rainfall depth of 41.4mm

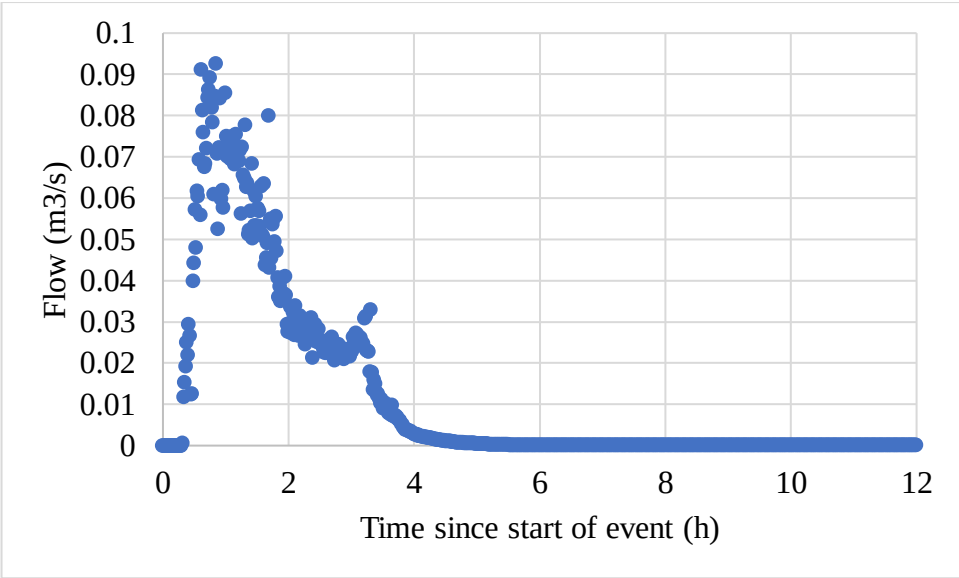

Figure 31. Overflow over time for a storm with total rainfall depth of 39.2mm

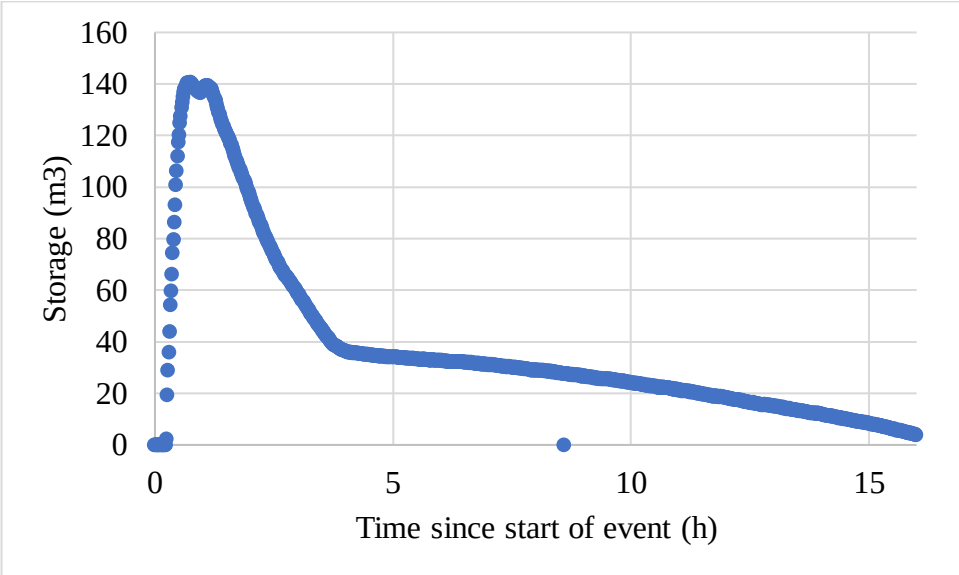

Figure 32. Surface storage over time for a storm with total rainfall depth of 39.2mm

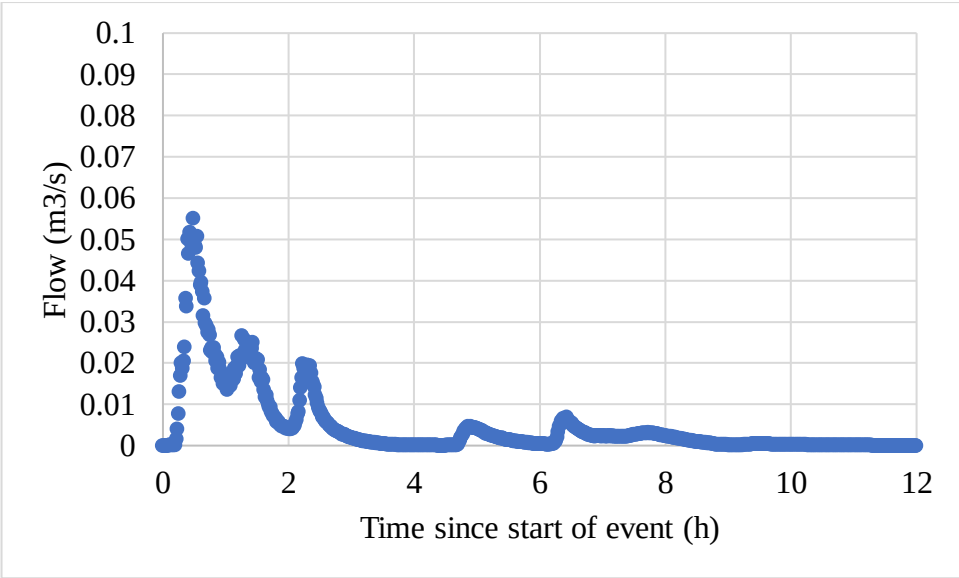

Figure 33. Outflow over time for a storm with total rainfall depth of 27.6mm

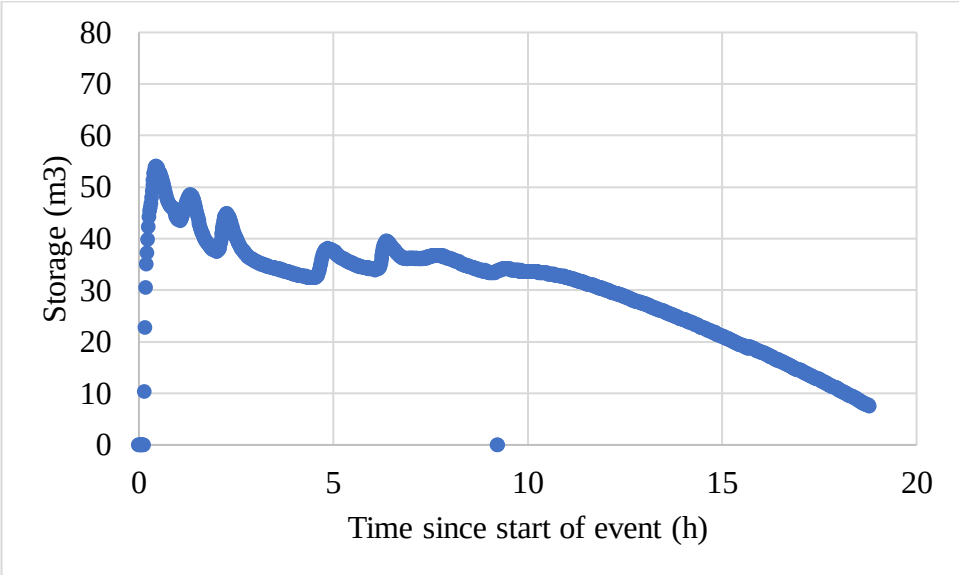

Figure 34. Surface storage over time for a storm with total rainfall depth of 27.6mm

7.2 Brazil

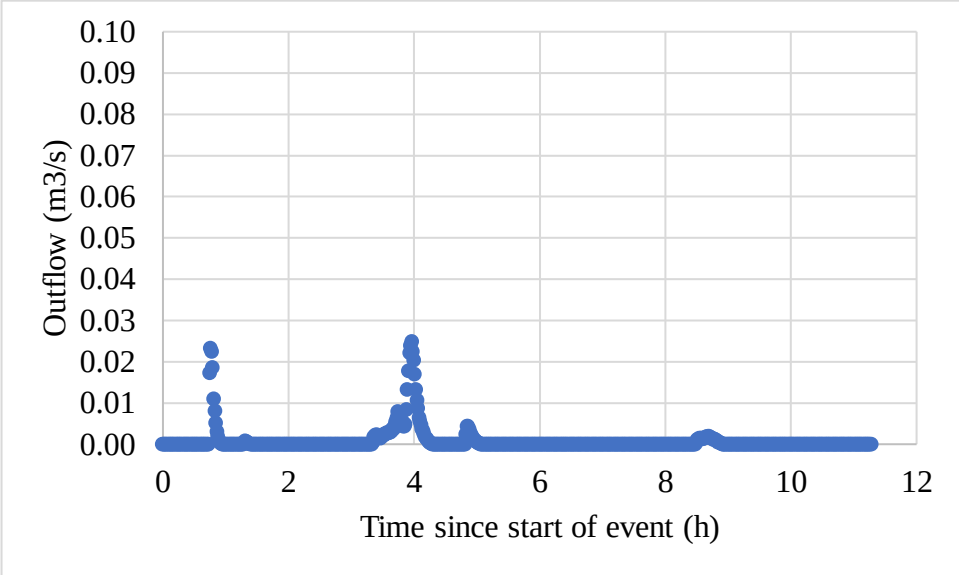

Figure 35. Overflow over time for a storm with total rainfall depth of 18.6mm

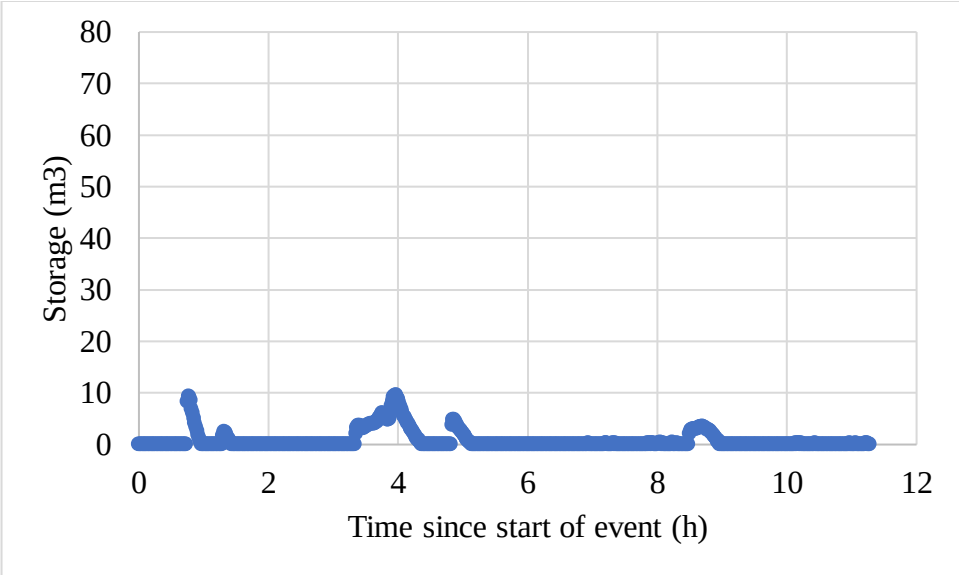

Figure 36. Surface storage over time for a storm with total rainfall depth of 18.6mm

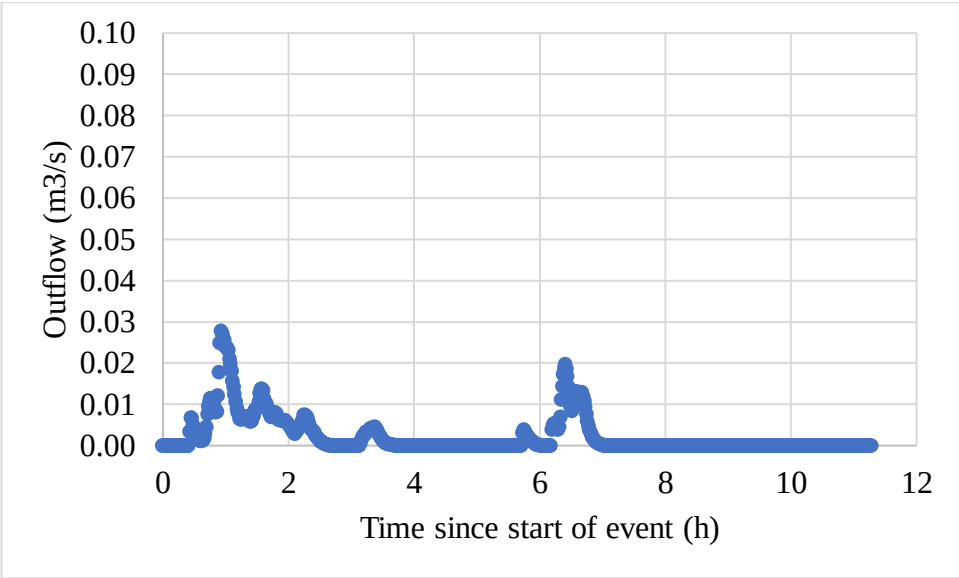

Figure 37. Overflow over time for a storm with total rainfall depth of 23.4mm

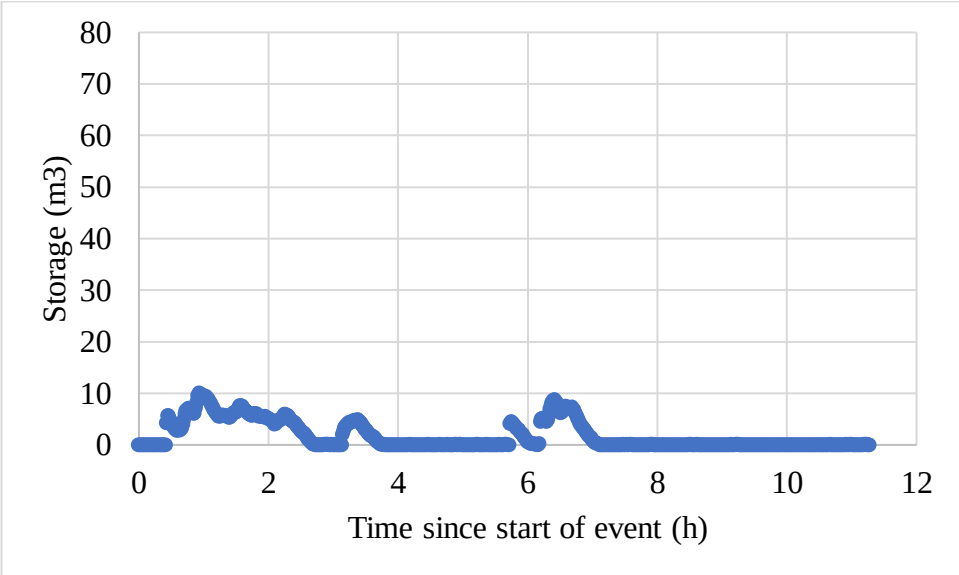

Figure 38. Surface storage over time for a storm with total rainfall depth of 23.4mm

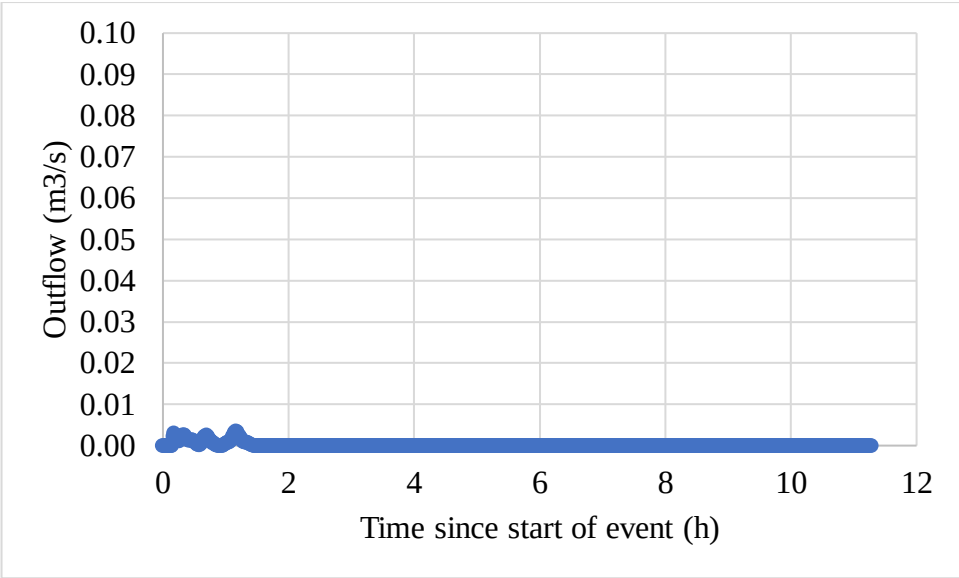

Figure 39. Overflow over time for a storm with total rainfall depth of 10mm

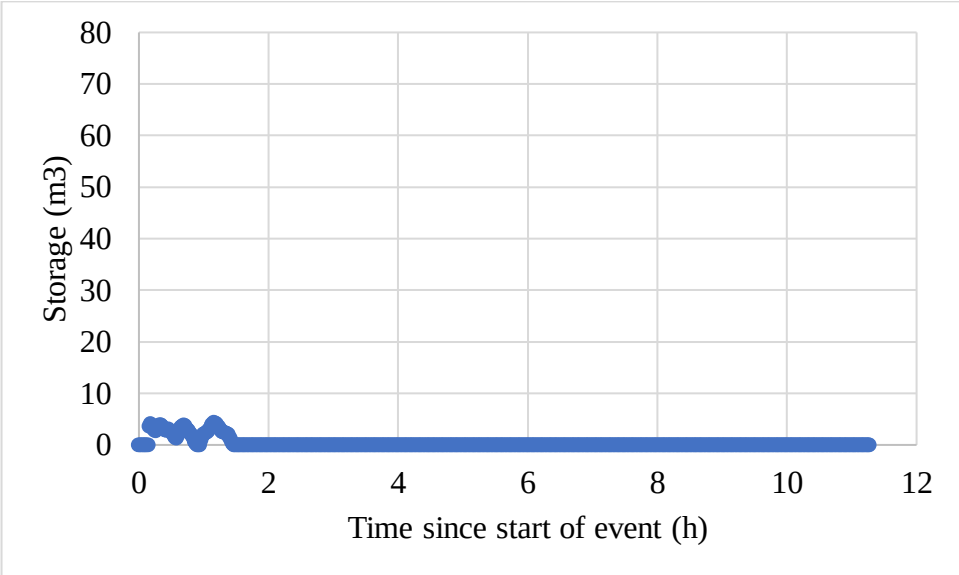

Figure 40. Surface storage over time for a storm with total rainfall depth of 10mm

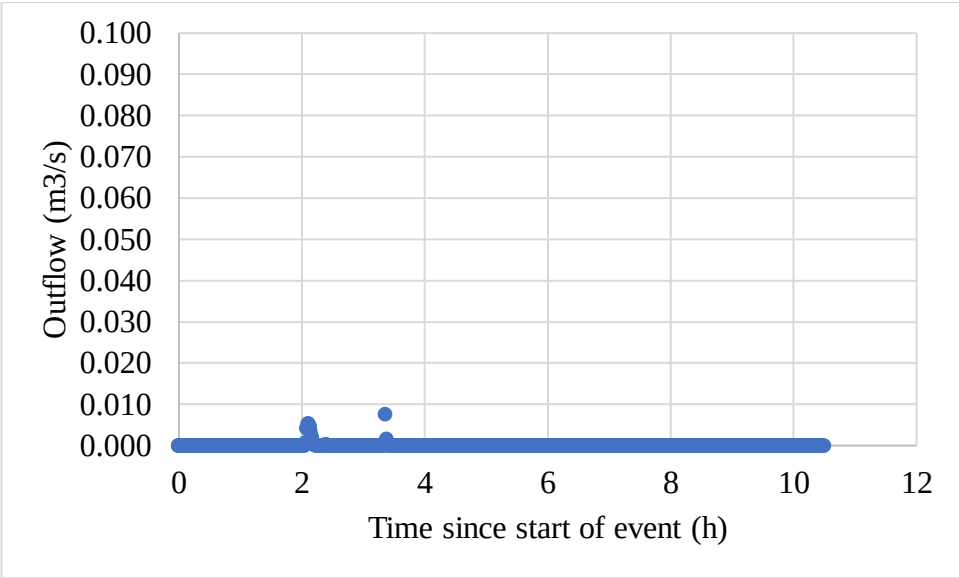

Figure 41. Overflow over time for a storm with total rainfall depth of 38.4mm

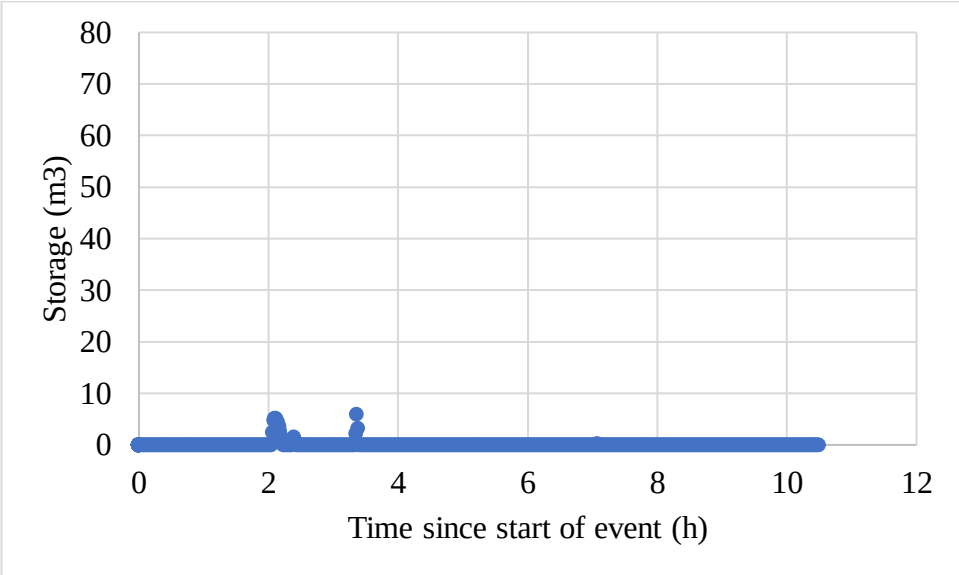

Figure 42. Surface storage over time for a storm with total rainfall depth of 38.4mm

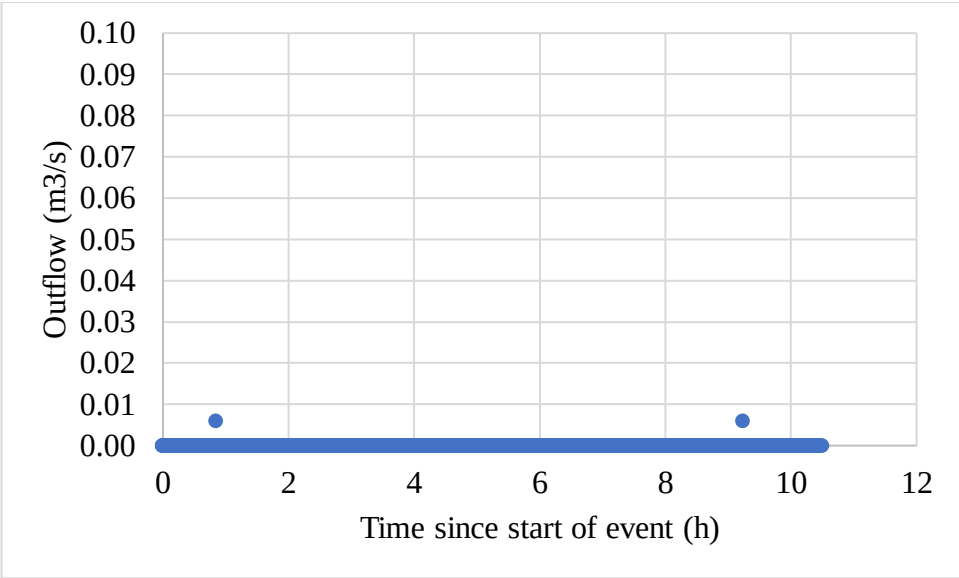

Figure 43. Overflow over time for a storm with total rainfall depth of 12.4mm

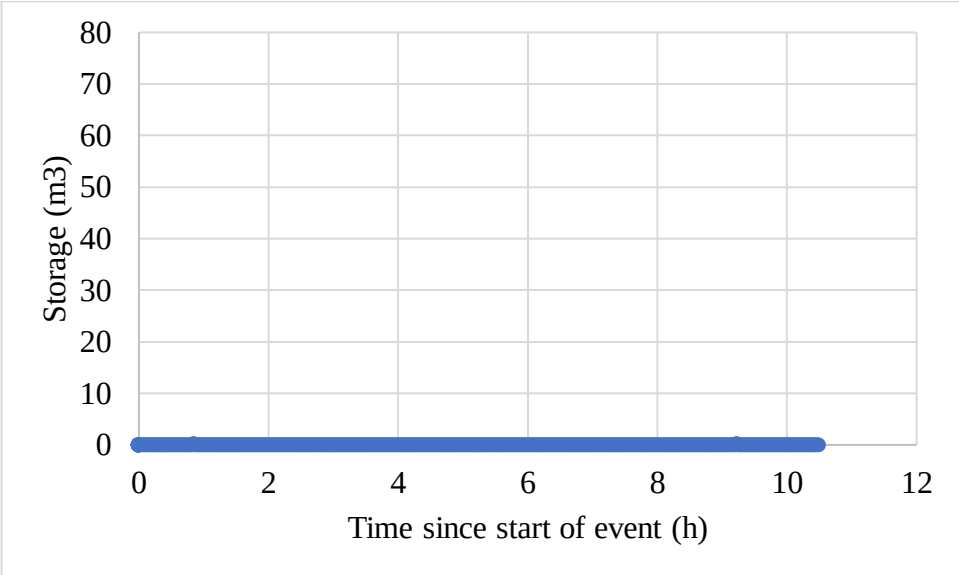

Figure 44. Surface storage over time for a storm with total rainfall depth of 12.4mm